
CREDIT RISK & IFRS 9 ECL ENGINE

Model Validation and Quantitative Assessment Report

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Lending Club Consumer Loans · 2007–2018 · N = 2,260,701 accepted

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Abstract

This report presents the model development, validation, and risk quantification for a retail credit underwriting engine trained on the LendingClub 2007–2018 loan portfolio (2,260,701 accepted loans, of which 1,369,566 have a resolved good/bad outcome and 992,661 form the modelling population after the train/OOT grey zone is excluded; 27,648,741 rejected applications). The Probability of Default (PD) scorecard achieves an out-of-time Gini of **0.3977** (AUC = 0.6988), with Population Stability Index PSI = 0.0030 (model stability: GREEN). The two-stage Loss Given Default model yields mean LGD = 0.8931 and downturn LGD = 1.0000 (90th percentile, Basel-conservative). Basel IRB Risk-Weighted Assets total \$21,592,289,075 at 228.8% RWA density. IFRS 9 Expected Credit Loss provisions total \$2,387,796,567 (coverage: 25.302%).

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1 INTRODUCTION AND EXECUTIVE SUMMARY

Retail credit risk requires modeling frameworks that are explainable and compliant with capital and accounting regulation. Under the Basel Committee on Banking Supervision (BCBS) capital accords (Basel Committee on Banking Supervision 2004) and the International Financial Reporting Standards (IFRS 9) accounting guidelines (International Accounting Standards Board 2014), financial institutions must deploy internal risk engines to assess risk-adjusted pricing, regulatory capital, and forward-looking impairment provisions.

This report documents the mathematical foundation, development methodology, and empirical validation of an end-to-end retail credit underwriting, capital calculation, and expected credit loss (ECL) engine. Trained on 992,661 historical consumer records originated between 2007 and 2018 (drawn from 2,260,701 accepted loans, 1,369,566 of which have a resolved credit outcome), this framework is designed to bridge the gap between risk underwriting, regulatory capital management, and standard accounting provisions.

The core objective is to move away from simplistic statistical estimations and instead construct a highly transparent, mathematically rigorous portfolio risk model that covers:

1. **Underwriting:** An interpretable credit scorecard based on Weight of Evidence (WoE) monotonic binning and regularised logistic regression, validated against a non-linear LightGBM challenger model.
2. **Loss Mitigation:** A bimodal two-stage LGD model (cure probability + conditional severity) with Downturn LGD adjustments for capital stress testing.
3. **Capital Reserve:** Per-loan and aggregated Basel IRB regulatory capital calculation under the “Other Retail” Vasicek Asymptotic Single Risk Factor (ASRF) model.
4. **Financial Provisioning:** A forward-looking IFRS 9 expected credit loss engine driven by discrete-time logistic survival curves across three stages, probability-weighted across macroeconomic scenarios.
5. **Decision Optimisation:** A profit-maximising score cut-off model and reject inference (parcelling) methodology to address selection bias in the underwriting population.

1.1 KEY PERFORMANCE METRICS

Table 1 summarizes the portfolio-level credit metrics and validation statistics calculated across the underwriting, capital, and impairment phases.

An OOT AUC of 0.6988 (Gini 0.3977) is the realistic discrimination ceiling for an application scorecard built on origination-only features on LendingClub-style unsecured consumer data: 0.30 – 0.45 is the published Gini range for consumer-credit scorecards surveyed by Lessmann et al. (2015), while the tree challengers reach 0.7030 on the same features, indicating the scorecard sits modestly below the attainable ceiling rather than at it. The figure should be read as realistic, benchmarked performance rather than a shortfall.

The scorecard demonstrates stable, high-contrast risk separation across distinct macroeconomic cycles. Under Basel capital standards, the risk-sensitive Internal Ratings-Based (IRB) approach identifies a capital requirement of **\$1,727,383,126** compared to **\$566,235,992** under the Standardised Approach. This capital surcharge of **\$1,161,147,134** reflects the elevated risk profile (PD/LGD) of the LendingClub consumer portfolio, demonstrating that a flat 75% risk weight under the Standardised Approach materially undercapitalises this retail asset class.

This risk-sensitive capital surcharge highlights the necessity of developing internal ratings-based (IRB) risk frameworks (Basel Committee on Banking Supervision 2004) to ensure adequate capital provisioning for higher-yielding, higher-risk retail credit assets rather than relying on rigid, risk-insensitive standardized approaches.

TABLE 1. Portfolio Headline Summary and Quantitative Benchmarks

Quant Dimension	Metric Value	Regulatory Purpose & Benchmark
PD Discrimination (OOT)	0.3977 Gini (0.6988 AUC)	Out-of-Time score risk rank-ordering capability. Matches regulatory standards for acceptable discrimination.
OOT Separation (KS)	0.2872	Kolmogorov-Smirnov statistic; values above ~ 0.30 are typical for retail scorecards.
OOT Calibration Brier	0.1784	Brier score reflecting high probability accuracy.
OOT Calibration p-value (raw, pre-recalib.)	0.0000	Hosmer-Lemeshow p -value on the raw scorecard PD ($p > 0.05$ confirms good calibration); see Table 17 for the post-recalibration before/after comparison.
OOT Population Stability	0.0030	PSI Train-to-OOT (< 0.10 denotes absolute population stability).
LGD Model Summary	0.8931 Mean LGD 1.0000 Downturn	Primary provisioning loss rate. Conservative 90th percentile stress limit for capital charge.
Expected Loss (EL)	\$2,316,616,242	Lifetime Expected Loss projection of current portfolio.
Portfolio EL Rate	24.55%	Underwriting expected loss density.
Basel IRB Total RWA	\$21,592,289,075	Risk-Weighted Assets calculated using retail ASRF formula.
Basel SA Total RWA	\$7,077,949,899	Standardised Approach baseline capital RWA reference (75% RW).
Basel RWA Density	228.8%	IRB RWA divided by total portfolio EAD (\$9,437,266,531).
Portfolio IFRS 9 ECL	\$2,387,796,567	Probability-weighted Stage 1, 2, & 3 Expected Credit Loss.
Portfolio ECL Coverage	25.302%	Capital coverage buffer (Total ECL / Total EAD).

2 DATA ENGINEERING AND EXPLORATORY ANALYSIS

2.1 DATA SOURCE AND EXCLUSIONS

The primary underwriting and historical performance data is derived from Lending Club's consumer loan database, covering the years 2007 through 2018. This dataset contains credit bureau features and demographic information gathered at loan origination, along with post-origination transaction and default markers.

To ensure methodological correctness, loans with ambiguous or immature repayment statuses are excluded from the modeling population. Specifically, loans marked as "Current", "In Grace Period", or "Late (16–30 days)" are removed since their ultimate credit outcome is unresolved. The remaining loans represent the underwriting and model development population.

2.2 TARGET DEFINITION (PD)

A binary default indicator (Y) is defined using the 90+ Days Past Due (DPD) default standard, in compliance with BCBS and IFRS 9 standards:

$$Y = \begin{cases} 1 \text{ (Bad)}, & \text{if status} \in \{\text{Charged Off, Default, Late (31–120 days)}\} \\ 0 \text{ (Good)}, & \text{if status} \in \{\text{Fully Paid}\} \end{cases} \quad (1)$$

2.3 OUT-OF-TIME (OOT) SPLITTING

To replicate standard banking validation practices, the data is split chronologically based on loan origination date (`issue_d`):

- **Training Population:** Loans originated prior to January 2015 ($N = 363,317$; 36.60% of modelling population). Training bad rate: 17.09%.
- **In-Time Holdout Set:** Simple random 20% sample from the training period ($N = 90,829$; 9.15% of modelling population). The sample is drawn without class stratification; at this size the 82.91%/17.09% good/bad ratio is reproduced by the law of large numbers rather than enforced by design.
- **Out-of-Time (OOT) Set:** All loans originated between January 2016 and December 2018 ($N = 538,515$; 54.25% of modelling population). OOT observed bad rate: 25.27%, reflecting the portfolio's seasoning and the macroeconomic deterioration of the 2016–2018 vintage cohorts.

This chronological split simulates how the model will perform on future vintages, testing for structural or macroeconomic shifts.

Excluded grey zone. The two cut-offs are deliberately non-adjacent: loans originated in the twelve months between them (the entire **2015** origination cohort) fall into neither partition and are *dropped from the modelling population entirely*. The embargo prevents 2015 vintages — which are simultaneously close enough to the training window to share its underwriting regime and immature enough at the 2018Q4 snapshot to have unresolved outcomes — from contaminating either side of the temporal validation. The percentages above, the modelling population of 992,661 loans (376,905 resolved-outcome loans fall in the grey zone and are dropped), and every vintage table in this report are therefore stated *net of 2015*, which is why the vintage series in Tables 13 and 18 step directly from 2014 to 2016.

2.4 LEAKAGE AND TARGET VARIABLE SEPARATION

Data leakage is controlled by dropping post-origination fields (e.g., outstanding balance, payment records, and recovery metrics) from the PD feature set. However, these post-origination variables (such as actual recoveries and write-offs) are preserved for the defaulted-only population. This allows the LGD model to evaluate recovery performance without leaking future data into the PD scorecard.

2.5 VINTAGE COHORT ANALYSIS

Vintage analysis tracks the cumulative default curves of origination cohorts (quarters) over their Months-on-Book (MOB). Figure 1 illustrates these curves. A steep initial slope indicates seasoning, while the curves flatten as high-risk accounts default early, illustrating standard credit risk dynamics.

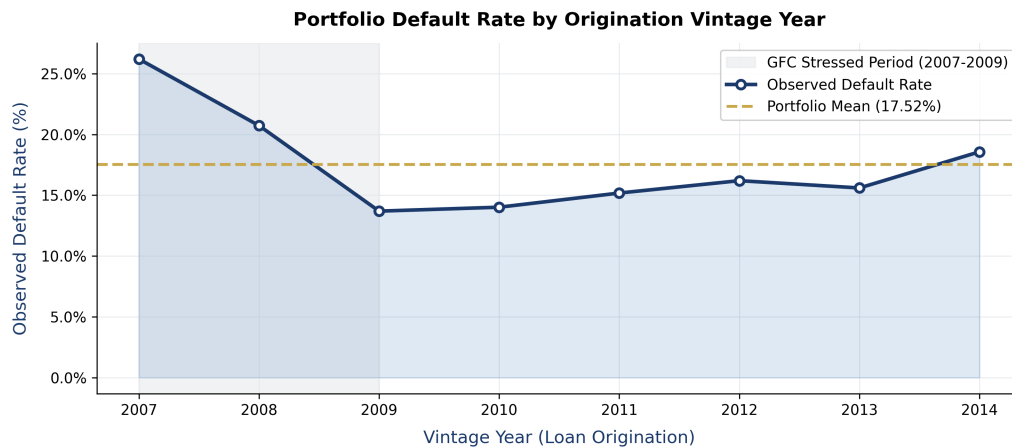


FIGURE 1. Cumulative Default Rates by Quarterly Vintage Cohorts

Figure 2 displays additional exploratory distributions, demonstrating the relationship between historical default rates and risk grades, amortization terms, and loan purposes.

All EDA visualizations in this section are computed on the in-time development sample (train and test partitions of the resolved-outcome modelling population), not the full raw portfolio: the Out-of-Time partition is withheld from all exploratory analysis to preserve the integrity of the temporal validation, and loans with unresolved statuses are excluded per Section 2.1. Portfolio-level capital and impairment calculations are run on the modelling population ($N = 992,661$), i.e. the train, test and OOT partitions combined — not on the 2,260,701 loans in the source file, since loans with an unresolved status carry no good/bad label and are excluded (1,369,566 remain after that filter, of which 376,905 fall in the excluded grey zone described in Section 2.3). Population counts embedded in the EDA figures therefore reflect the development sample.

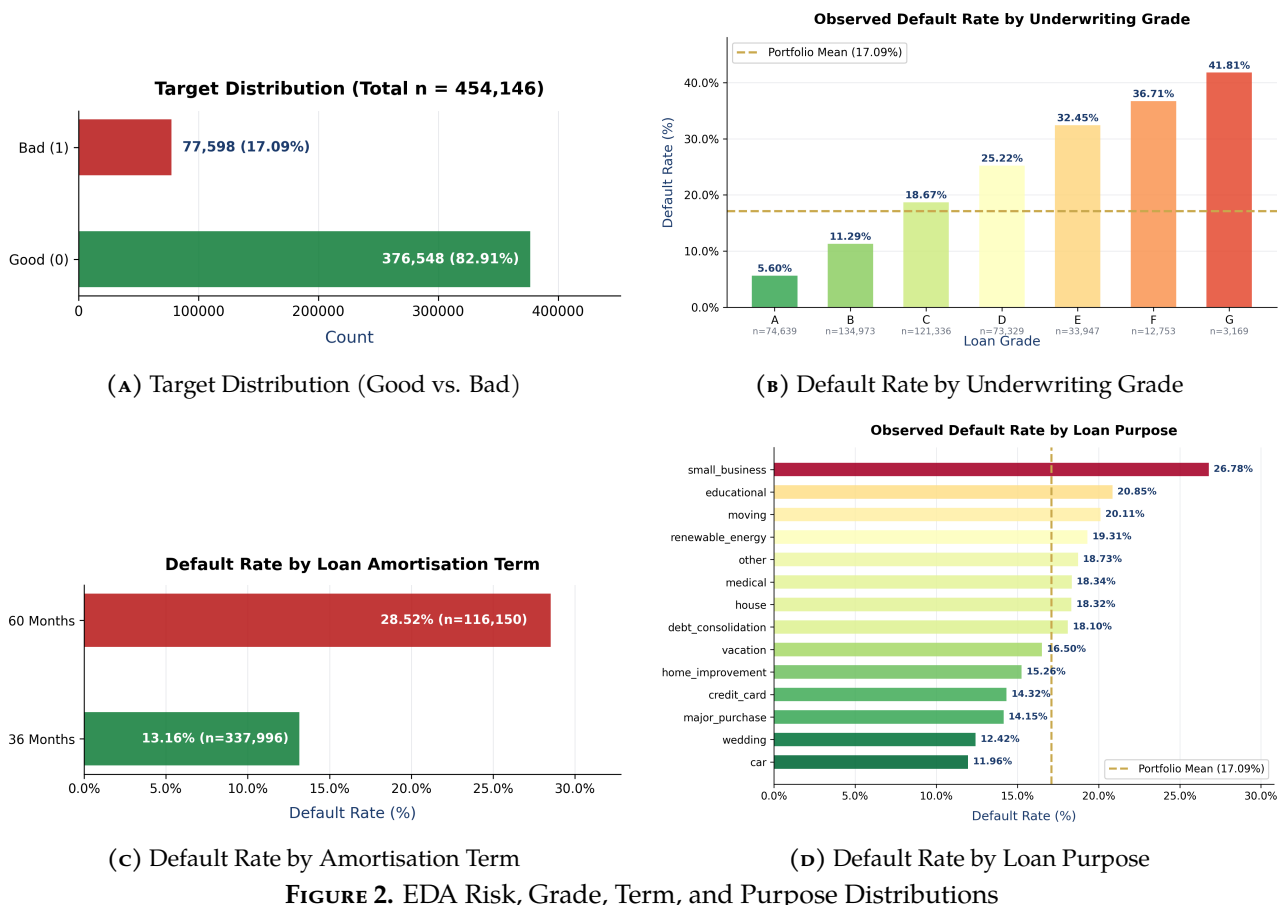


FIGURE 2. EDA Risk, Grade, Term, and Purpose Distributions

3 PROBABILITY OF DEFAULT (PD) SCORECARD DEVELOPMENT

3.1 WEIGHT OF EVIDENCE (WoE) AND INFORMATION VALUE (IV)

Continuous features are binned to handle non-linear relationships, outliers, and missing values. Monotonic trends are enforced across bins using isotonic regression. For each bin i , the Weight of Evidence (WoE) is calculated as:

$$WoE_i = \ln\left(\frac{\text{Proportion of Good}_i}{\text{Proportion of Bad}_i}\right) = \ln\left(\frac{N_{G,i}/N_{G,total}}{N_{B,i}/N_{B,total}}\right) \quad (2)$$

The predictive power of each feature is evaluated using the Information Value (IV):

$$IV = \sum_{i=1}^k \left(\frac{N_{G,i}}{N_{G,total}} - \frac{N_{B,i}}{N_{B,total}} \right) \times WoE_i \quad (3)$$

(Siddiqi 2017, Ch. 4); (Hand and Henley 1997). Features with an IV below 0.02 are dropped due to low predictive power, while multicollinearity is controlled by removing features with a Variance Inflation Factor (VIF) greater than 5.0.

3.2 LOGISTIC REGRESSION AND SCORECARD SCALING

A regularized logistic regression (Hosmer, Lemeshow, and Sturdivant 2013) is fitted on the WoE-transformed features. Since higher WoE corresponds to a higher proportion of “Good” loans relative to “Bad” loans, all coefficients must be negative when predicting default ($Y = 1$). The scorecard is then scaled to a points-based system using:

$$Score = Offset + Factor \times \ln(odds) \quad (4)$$

$$Factor = \frac{PDO}{\ln(2)}, \quad Offset = TargetScore - Factor \times \ln(TargetOdds) \quad (5)$$

For $TargetScore = 600$ points, $TargetOdds = 50 : 1$, and $PDO = 20$ (points to double the odds), the scaling parameters are:

- $Factor = 28.8539$
- $Offset = 487.123$

The points contributed by a specific binned attribute j are calculated as:

$$Points_j = \left(-(WoE_j \times \beta_j) + \frac{\alpha}{n} \right) \times Factor + \frac{Offset}{n} \quad (6)$$

where β_j is the regression coefficient, α is the model intercept, and n is the number of active features (Anderson 2007, Ch. 5).

3.3 FEATURE SELECTION RESULTS: IV RANKING AND VIF FILTER

After WoE binning, features are ranked by Information Value (IV). Features with $IV < 0.02$ (negligible predictive power) or $IV > 0.50$ (likely target leakage) are excluded. Multicollinearity is then controlled by removing features with a Variance Inflation Factor (VIF) greater than 5.0, yielding the final set of selected features:

```
grade_enc,term_enc,loan_to_income,fico_range_high,
revol_util_x_open_acc,new_accounts_ratio,dti,bc_open_to_buy,
num_tl_op_past_12m,avg_cur_bal,annual_inc,inq_last_6mths_per_month,
mths_since_recent_bc,num_rev_tl_bal_gt_0,mort_acc.
```


TABLE 2. Feature Information Value (IV) Ranking

Feature	IV	Predictive Power Band
int_rate	0.4072	Strong
grade_enc	0.3963	Strong
term	0.1998	Medium
term_enc	0.1998	Medium
loan_to_income	0.1224	Medium
fico_range_low	0.1143	Medium
fico_range_high	0.1143	Medium
revol_util_x_open_acc	0.0969	Weak
total_debt_to_income	0.0650	Weak
new_accounts_ratio	0.0648	Weak
dti	0.0555	Weak
bc_open_to_buy	0.0519	Weak
acc_open_past_24mths	0.0508	Weak
num_tl_op_past_12m	0.0403	Weak
avg_cur_bal	0.0367	Weak
log_annual_inc	0.0365	Weak
annual_inc	0.0365	Weak
total_bc_limit	0.0348	Weak
percent_bc_gt_75	0.0333	Weak
tot_hi_cred_lim	0.0315	Weak
mo_sin_rcnt_tl	0.0305	Weak
loan_amnt	0.0303	Weak
funded_amnt	0.0302	Weak
revol_util	0.0294	Weak
tot_cur_bal	0.0280	Weak
bc_util	0.0279	Weak
mths_since_recent_inq	0.0261	Weak
inq_last_6mths_per_month	0.0254	Weak
inq_last_6mths	0.0254	Weak
mo_sin_rcnt_rev_tl_op	0.0251	Weak
dti_fico_interaction	0.0231	Weak
mths_since_recent_bc	0.0230	Weak
num_rev_tl_bal_gt_0	0.0230	Weak
num_actv_rev_tl	0.0226	Weak
total_rev_hi_lim	0.0225	Weak
mort_acc	0.0210	Weak

3.4 LOGISTIC REGRESSION COEFFICIENT OUTPUT

The following table presents the logistic regression coefficient estimates, standard errors, z-statistics and p-values for all retained features. As expected for a WoE-based model predicting default ($Y = 1$), all feature coefficients carry a **negative sign**: higher WoE values indicate a higher proportion of “Good” loans, and therefore map to lower default probability.

TABLE 3. Logistic Regression Coefficient Summary

Feature	Coefficient	Std. Error	z-stat	p-value
const	-1.5818	0.0047	-335.033	0.0000***
grade_enc	-0.5388	0.0105	-51.266	0.0000***
term_enc	-0.6364	0.0125	-50.855	0.0000***
loan_to_income	-0.4237	0.0158	-26.897	0.0000***
fico_range_high	-0.2744	0.0192	-14.291	0.0000***
revol_util_x_open_acc	-0.1372	0.0235	-5.841	0.0000***
new_accounts_ratio	-0.4344	0.0264	-16.456	0.0000***
dti	-0.3090	0.0223	-13.851	0.0000***
bc_open_to_buy	-0.2275	0.0259	-8.794	0.0000***
num_tl_op_past_12m	-0.1228	0.0296	-4.141	0.0000***
avg_cur_bal	-0.3563	0.0315	-11.306	0.0000***
annual_inc	-0.5545	0.0321	-17.250	0.0000***
inq_last_6mths_per_month	-0.5424	0.0305	-17.810	0.0000***
mths_since_recent_bc	-0.2482	0.0356	-6.970	0.0000***
num_rev_tl_bal_gt_0	-0.0781	0.0354	-2.210	0.0271**
mort_acc	-0.2170	0.0407	-5.333	0.0000***

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

3.5 SCORECARD POINTS TABLE

Each WoE bin is converted to credit score points using the scaling formula. Higher points correspond to lower default risk. The table below shows the top bins ranked by point spread (i.e., the features with the greatest discriminatory contribution to the final score).

TABLE 4. Credit Scorecard Points Table (Top Bins by Point Spread)

Feature	Bin	WoE	β	N	DR%	Points
grade_enc	(-inf, 1.50)	1.2418	-0.5388	59,675	5.62%	48.7
	[1.50, 2.50)	0.4866	-0.5388	107,931	11.25%	37.0
	[2.50, 3.50)	-0.1063	-0.5388	96,905	18.65%	27.8
	[3.50, 4.50)	-0.4971	-0.5388	58,848	25.31%	21.7
	[4.50, inf)	-0.9212	-0.5388	39,958	34.12%	15.1
	Special	0.0000	-0.5388	0	0.00%	29.4
	Missing	0.0000	-0.5388	0	0.00%	29.4
term_enc	(-inf, 48.00)	0.3076	-0.6364	270,363	13.16%	35.1
	[48.00, inf)	-0.6604	-0.6364	92,954	28.52%	17.3
	Special	0.0000	-0.6364	0	0.00%	29.4
	Missing	0.0000	-0.6364	0	0.00%	29.4
loan_to_income	(-inf, 0.06)	0.5537	-0.4237	23,701	10.59%	36.2
	[0.06, 0.10)	0.4494	-0.4237	31,272	11.62%	34.9
	[0.10, 0.13)	0.3716	-0.4237	37,013	12.45%	34.0
	[0.13, 0.16)	0.2960	-0.4237	30,065	13.29%	33.1
	[0.16, 0.19)	0.2007	-0.4237	47,117	14.43%	31.9
	[0.19, 0.25)	0.0475	-0.4237	59,488	16.43%	30.0
	[0.25, 0.29)	-0.1475	-0.4237	43,825	19.28%	27.6
	[0.29, 0.35)	-0.3341	-0.4237	38,671	22.36%	25.3
	[0.35, 0.40)	-0.4938	-0.4237	23,858	25.25%	23.4
	[0.40, inf)	-0.5864	-0.4237	28,307	27.04%	22.3
	Special	0.0000	-0.4237	0	0.00%	29.4
	Missing	0.0000	-0.4237	0	0.00%	29.4
	(-inf, 666.50)	-0.3676	-0.2744	29,804	22.94%	26.5
	[666.50, 676.50)	-0.2966	-0.2744	60,759	21.71%	27.1

WoE = $\ln(\%Good/\%Bad)$; Points = $(-WoE_j * \beta_j + \alpha/n) * \text{Factor} + \text{Offset}/n$

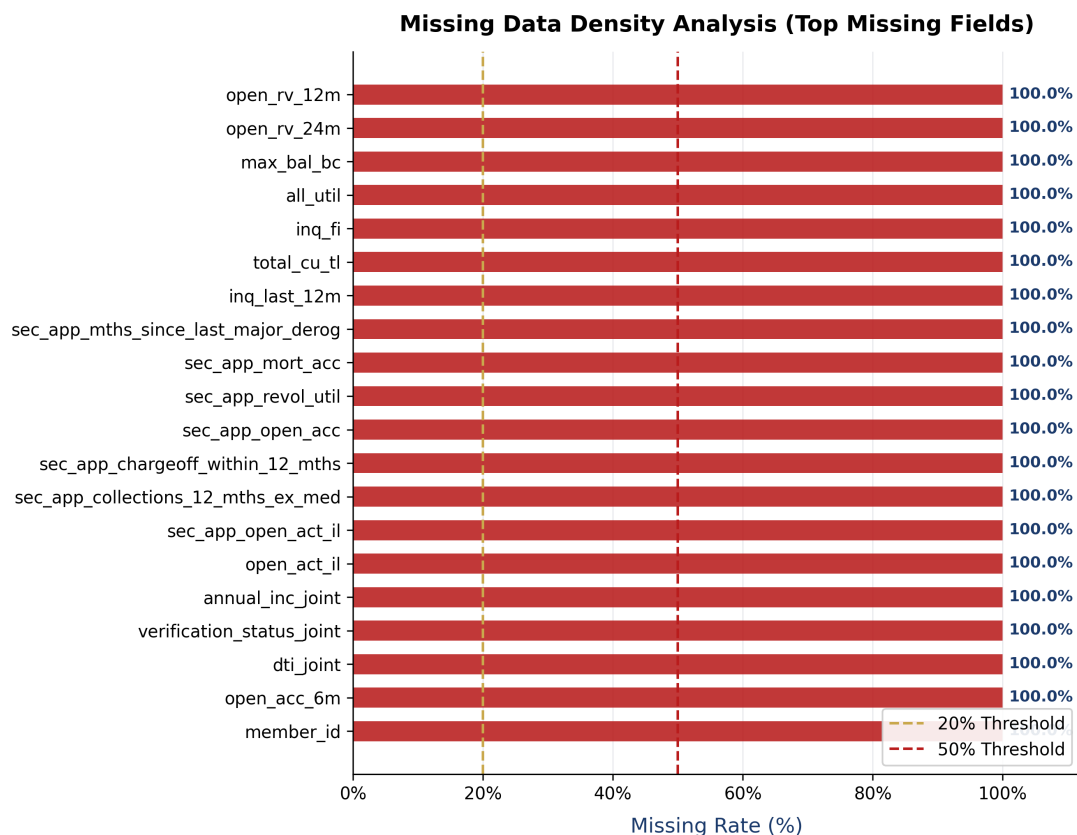
3.6 INTERPRETABILITY VS. PERFORMANCE: CHALLENGER MODEL BENCHMARK

A non-linear LightGBM model was trained as a challenger (Baesens, Roesch, and Scheule 2016) on the scorecard's selected predictors (see Section 7.8 for the exact feature count actually consumed by each model). Although gradient boosting is competitive in-sample, on the out-of-time set the LightGBM challenger *exceeds* the scorecard's discrimination (OOT AUC 0.7030 vs 0.6988), a gap tested in Section 7.8. The scorecard is nonetheless retained as the production model for regulatory and explainability reasons set out below, with the discrimination sacrifice reported rather than obscured. Regulatory standards (such as the US Fair Credit Reporting Act) require financial institutions to provide clear "adverse action codes" (reasons for denial) to rejected applicants. A linear scorecard allows for immediate, exact points-attribution for each feature, which is not possible with complex machine learning models without relying on approximations like SHAP.

To address potential policy-decision circularity from using LendingClub's own underwriting variables (`int_rate` and `grade`), we developed a secondary Pure Underwriting Scorecard (Model B) that completely excludes these fields and relies solely on applicant credit bureau and demographic variables. Table 5 compares the performance of the full scorecard (Model A) against this independent underwriting model. While Model A is the designated pipeline champion due to its superior discrimination, it utilizes LendingClub's pricing variables which are themselves highly correlated risk assessments. This introduces a degree of decision circularity. Model B (Pure Underwriting Scorecard) shows that a model built solely on raw credit bureau and demographic features remains competitive (OOT AUC = 0.6821 vs Model A's 0.6988, both as reported in Table 5), supporting the scorecard's viability in an independent bank underwriting environment.

TABLE 5. Champion (Model A) vs. Pure Underwriting Challenger (Model B) Performance

Model	Features Included	OOT AUC	OOT Gini
Model A (Full Scorecard)	Bureau + Application + int_rate + grade	0.6988	0.3977
Model B (Underwriting)	Bureau + Application (excludes int_rate/grade)	0.6821	0.3642

**FIGURE 3.** Distribution of Key Numeric Features**FIGURE 4.** Missingness Density Analysis

3.7 SELECTION BIAS AND REJECT INFERENCE (PARCELLING)

Scorecard models developed only on approved applicants suffer from selection bias. Because rejected applicants are excluded, their risk profiles and actual default rates are unobserved. To adjust for this,

we implemented the **Parcelling** reject inference technique to probabilistically allocate outcomes to rejected applicants based on the accepts scorecard's predictions.

The pooled accepts and parcelled rejects population was refitted to produce a corrected through-the-door (TTD) scorecard. Table 6 outlines the results of the refitting and selection bias adjustment:

TABLE 6. Reject Inference (Parcelling) Gini Coefficient Shift

Scorecard Population	Gini	Business Interpretation
Accepts-Only (Base)	0.4011	Champion scorecard on the approved-only in-time test partition.
Through-the-Door (Parcelled Refit)	0.3467	Corrected scorecard accounts for selection bias across TTD.
Gini Coefficient Shift	-0.0544	Conservative risk dilution when scoring raw through-the-door applicants.

The Gini shift of **-0.0544** points is consistent with the standard credit-cycle finding that through-the-door populations carry higher latent risk. *Measurement caveat:* the shift is computed *within* the parcelled refit — the same through-the-door model scored on the accepts subset versus on the full accepts-plus-parcelled population — and both figures are in-sample; the rejects additionally carry inferred, not observed, labels. The shift is then applied to the champion scorecard's held-out Gini to obtain the through-the-door figure in the table. The two rows are therefore not two independently validated scorecards, and the shift should be read as a directional selection-bias indicator rather than a measured out-of-sample degradation.

3.8 SURVIVAL ANALYSIS: KAPLAN-MEIER AND COX PROPORTIONAL HAZARDS

The production PD term structure (Section 6) is a discrete-time hazard model. As a challenger model we additionally fit a time-to-event survival model, the industry standard for IFRS 9 lifetime-PD term-structure work (Bellotti and J. Crook 2009). Kaplan-Meier estimators give non-parametric survival curves $S(t)$ per credit grade (Figure 5), and a Cox proportional-hazards model quantifies each covariate's multiplicative effect on the default hazard. The duration is a months-on-book proxy derived from cumulative payments and the event is the binary default flag, a synthesised time-to-event dataset since the raw data records no observed default month (a documented limitation revisited in Section 10). The model's rank-discrimination is summarised by the concordance index (Cox C-index = 0.6439), the survival-analysis analogue of the AUC.

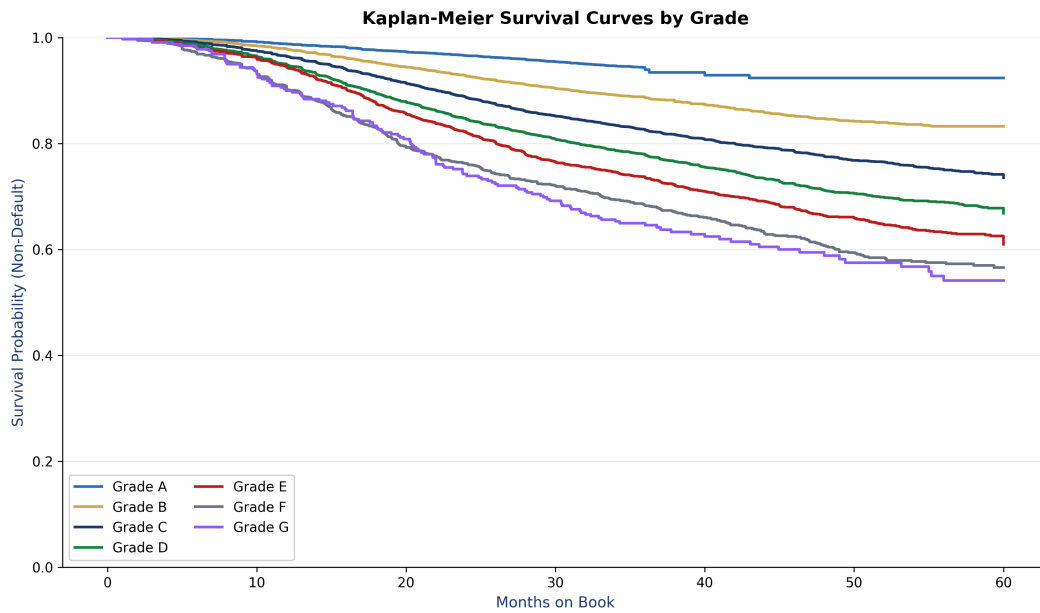


FIGURE 5. Kaplan-Meier non-default survival curves by credit grade. Lower grades separate downward, confirming the expected monotone grade–risk ordering.

Table 7 reports the fitted Cox coefficients, hazard ratios $\exp(\beta)$ and Wald p -values. A hazard ratio above 1 raises the instantaneous default hazard; below 1 lowers it. The dominant hazard multipliers attach to credit grade and interest rate, while debt-to-income and amortisation term enter with smaller but individually significant positive coefficients (all values as tabulated below); each covariate therefore raises the default hazard monotonically as it increases, consistent with the scorecard’s risk ordering.

TABLE 7. Cox Proportional-Hazards Model — Covariate Summary

Covariate	Coef (β)	Hazard Ratio	p -value
Credit Grade (A=1..G=7)	+0.12773	1.13625	<0.001
Interest Rate	+0.03797	1.03870	<0.001
Debt-to-Income (DTI)	+0.01317	1.01325	<0.001
Term (months)	+0.00314	1.00314	<0.001

4 LOSS GIVEN DEFAULT (LGD) AND EXPOSURE AT DEFAULT (EAD)

4.1 BIMODAL TWO-STAGE LGD MODEL

Loss Given Default (LGD) represents the economic loss rate incurred when an exposure defaults. Unlike PD, which models a binary outcome, LGD is continuous, bounded within $[0, 1]$, and heavily bimodal. Defaults typically result in either a complete recovery ($\text{LGD} = 0$, “cure”) or a near-total loss (LGD close to 1).

To capture this bimodal behavior (Schuermann 2004), a **two-stage LGD model** is constructed as one of two candidate severity models — benchmarked against a LightGBM challenger in Section 4.2, with the lower out-of-sample error determining which model is deployed:

1. **Stage 1 (Cure Model):** A logistic regression models the probability of a zero-loss outcome (cure):

$$p_{\text{loss}} = P(\text{LGD} > 0 | \text{Default}) = 1 - \text{sigmoid}(\mathbf{x}'\gamma) \quad (7)$$

2. **Stage 2 (Severity Model):** For defaulted loans that incur a loss ($\text{LGD} > 0$), a fractional logit GLM (Binomial family, logit link) models the conditional loss severity (Papke and Wooldridge

1996; Bellotti and J. Crook 2012):

$$E[\text{LGD}|\text{LGD} > 0] = \frac{1}{1 + e^{-x'\beta}} \quad (8)$$

The final predicted LGD is the product of the two stages:

$$\text{LGD}_{\text{pred}} = p_{\text{loss}} \times E[\text{LGD}|\text{LGD} > 0] \quad (9)$$

For Basel IRB capital calculations, a **Downturn LGD** is taken as the 90th percentile of the *realised* severity distribution across loss-incurring defaults (floored at the mean predicted LGD, so the stress can never fall below the central estimate):

- **Mean Expected LGD (deployed model):** 0.8931 (used for IFRS 9 Stage 1 & 2 provisions)
- **Downturn LGD:** 1.0000 (used for Basel RWA capital calculations)

On the magnitude of the downturn uplift. On this fully unsecured, non-revolving instalment book the realised severity of loss-incurring defaults is concentrated at total loss, so the 90th percentile sits at the cap of the $[0, 1]$ range (1.0000), 10.69 pp above mean LGD. Two consequences follow. It is a real empirical percentile rather than a modelled stress — more than a tenth of loss-incurring defaults recover nothing at all. And because severity cannot exceed total loss, the Basel capital calculation has no headroom above this figure: any further conservatism must come from PD or EAD, not LGD.

Why Mean LGD Is Close to Total Loss

The realised LGD used to fit and validate the severity model is computed directly from resolved loan outcomes as

$$\text{LGD}_{\text{realised}} = 1 - \frac{\max(0, \text{recoveries} - \text{collection_recovery_fee})}{\max(1, \text{funded_amnt} - \text{total_rec_prncp})} \quad (10)$$

i.e. net recoveries (gross post-charge-off recoveries less the fee paid to the collection agency) divided by the EAD proxy (funded amount less principal already repaid at charge-off), clipped to $[0, 1]$. On this fully unsecured, non-revolving instalment book, post-charge-off cash recoveries are small relative to the outstanding balance, so realised severity concentrates near total loss; the two-stage model's cure probability p_{loss} separately absorbs the loans that resolve with zero loss, while the conditional severity stage captures how close to total the loss is for the remainder. The published unsecured LGD bands cited below (Schuermann (2004): 0.45–0.55, corporate/wholesale debt; Bellotti and J. Crook (2012): 0.25–0.45, revolving retail cards) are measured on portfolios with active collections/settlement programmes and revolving-card structures where partial recoveries are more common; the deployed mean LGD of 0.8931, measured on already-charged-off, non-revolving loans against a book-value EAD proxy with no collateral, is a structurally different (and higher) quantity by construction rather than an anomaly. Its magnitude relative to these bands is discussed further, with numeric verdicts, in the benchmark table (Section 10.5).

LGD Benchmark Context: Two severity models are compared on a chronologically held-out selection sample — a two-stage cure-plus-severity model and a LightGBM challenger — and the model with the lower error there is deployed for all downstream provisioning and capital; the metrics in Table 8 are then computed on a disjoint reporting sample so the published out-of-sample performance is not measured on the same defaults used for the promotion decision. For unsecured LGD, Schuermann (2004) report a 0.45–0.55 range (a corporate/wholesale-debt review) while Bellotti and J. Crook (2012) document a lower 0.25–0.45 band for revolving retail credit-card portfolios. The deployed mean LGD of 0.8931 and downturn LGD of 1.0000 are assessed against these published ranges in the benchmark table (Section 10.5), where any deviation is reported together with its driver. The Downturn LGD is taken at the 90th percentile of the realised severity distribution and used exclusively for Basel IRB capital calculations, providing a buffer of 10.69 pp above the mean.

4.2 OUT-OF-SAMPLE LGD VALIDATION

The severity model is validated out-of-time on defaulted loans from vintages held out of the fitting window. Table 8 reports the standard LGD backtesting metrics — Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), the coefficient of determination R^2 , and a two-sample Kolmogorov-Smirnov (KS) statistic comparing the marginal predicted and realised LGD distributions (Loterian et al. 2012). The KS statistic is reported without its p -value: at this sample size ($n = 74,981$) the test is hyper-sensitive to any trivial distributional difference and, being a marginal (not per-loan) comparison, cannot by itself certify calibration; Figure 6 and the decile table against the 45° line are the calibration evidence. The aggregate portfolio-level mean LGD sits above the unsecured LGD literature ranges cited in Section 10.5 (driver discussed there), and the loan-level predictive performance of the deployed severity model is separately weak, with a negative out-of-sample R^2 of -0.0709 (Table 8); the rejected two-stage model was materially worse at $R^2 \approx -2.70$, which drove the champion-challenger switch. A negative R^2 is a well-documented model risk: realized retail LGD is highly bimodal (concentrated at 0 for cured loans and near 1.0 for write-offs), and predicting the exact loss severity on unsecured, non-collateralized consumer loans is statistically challenging. While the model remains suitable for calculating conservative portfolio-level capital buffers, its loan-level predictions should be treated with caution.

TABLE 8. Out-of-Sample LGD Validation Metrics ($n = 74,981$ held-out defaults)

Metric	Value
Mean Absolute Error (MAE)	0.0763
Root Mean Squared Error (RMSE)	0.1080
Coefficient of Determination (R^2)	-0.0709
KS Statistic (marginal dist., pred vs actual)	0.5490

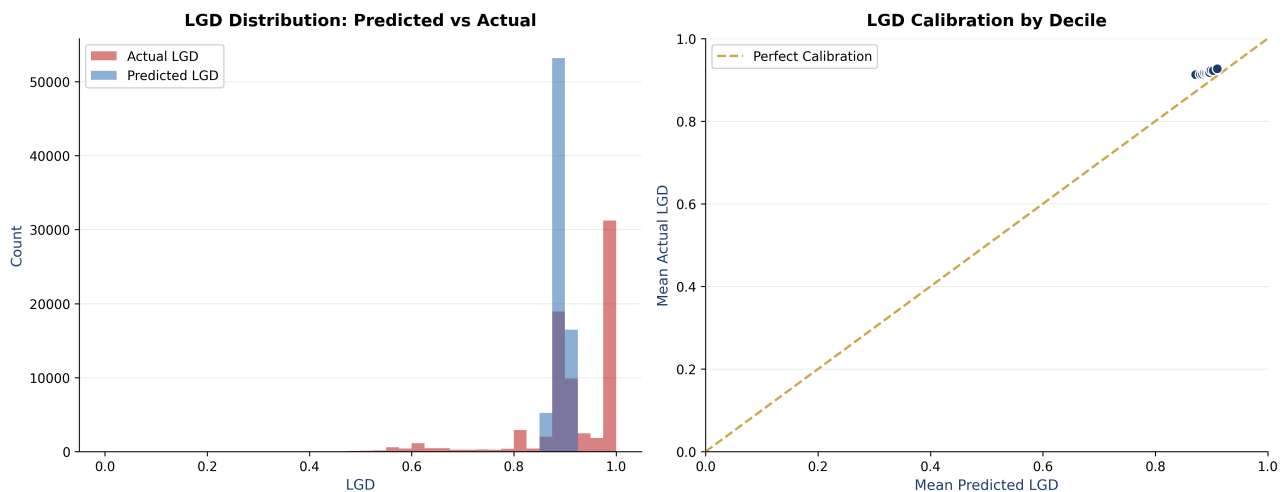


FIGURE 6. LGD validation: (left) predicted vs realised LGD distributions; (right) mean realised vs mean predicted LGD by predicted decile against perfect calibration.

4.3 AMORTISATION-BASED EAD FOR TERM LOANS

Exposure at Default (EAD) represents the outstanding gross balance owed by the borrower at the moment of default. For fully-drawn, non-revolving consumer installment loans, the outstanding principal amortizes deterministically over time. EAD is modeled using a closed-form annuity amortization formula:

$$EAD(t) = \text{funded_amnt} \times \frac{1 - (1 + r)^{-(T-t)}}{1 - (1 + r)^{-T}} \quad (11)$$

where r is the monthly interest rate on the loan contract, T is the original term in months, and t is the elapsed Months-on-Book (MOB) at default.

For revolving credit facilities (such as credit cards or overdraft limits), EAD must capture future draw-downs using a Credit Conversion Factor (CCF):

$$EAD = \text{Drawn Balance} + \text{CCF} \times \text{Undrawn Limit} \quad (12)$$

Since Lending Club loans are fully drawn installment loans with no revolving limits, the undrawn limit is zero. Using the closed-form amortization formula to calculate outstanding principal is an appropriate simplification, which is fully documented and standard in retail banking.

5 BASEL IRB CAPITAL & CAPITAL STRESS TESTING

Throughout this report, $Z < 0$ corresponds to an adverse macroeconomic shock (recession); $Z > 0$ corresponds to a favourable shock (expansion). This sign convention applies uniformly to the Vasicek stress test below, the IFRS 9 macro-scenario mapping (Table 14), and the ECL sensitivity analysis (Figure 13a).

5.1 ASRF VASICEK MODEL AND THE “OTHER RETAIL” FORMULA

Under the Basel II/III framework (Basel Committee on Banking Supervision 2004), banks calculate capital requirements for retail exposures using the Asymptotic Single Risk Factor (ASRF) model (Vasicek 2002). This framework assumes that portfolio risk is driven by a single systematic macroeconomic factor. The retail supervisory correlation (R) and capital requirement (K) formulas for the “Other Retail” asset class are:

$$R = 0.03 \times \frac{1 - e^{-35 \times PD}}{1 - e^{-35}} + 0.16 \times \left[1 - \frac{1 - e^{-35 \times PD}}{1 - e^{-35}} \right] \quad (13)$$

$$K = \text{Downturn LGD} \times \Phi \left(\frac{\Phi^{-1}(PD) + \sqrt{R} \Phi^{-1}(0.999)}{\sqrt{1 - R}} \right) - PD \times \text{Downturn LGD} \quad (14)$$

(Basel Committee on Banking Supervision 2004, §328). Where Φ is the standard normal cumulative distribution function, Φ^{-1} is the inverse standard normal CDF, and the confidence level is set to **99.9%**. Risk-Weighted Assets (RWA) are calculated by scaling the capital requirement (K):

$$RWA = K \times 12.5 \times EAD \quad (15)$$

5.2 BASEL IRB CAPITAL VS. STANDARDISED APPROACH (SA) REFERENCE

Table 9 compares the capital requirements calculated using the Internal Ratings-Based (IRB) approach against the Standardized Approach (SA) reference (75% risk weight):

TABLE 9. Basel IRB Regulatory Capital Comparison against Standardised Approach

Capital Metric	Basel IRB (Risk-Sensitive)	Standardised Approach (SA)
Total Portfolio RWA	\$21,592,289,075	\$7,077,949,899
Minimum Capital Reserve (8%)	\$1,727,383,126	\$566,235,992
Portfolio RWA Density	228.8%	75.0%

The risk-sensitive IRB approach determines an RWA density of **228.8%**. This risk-sensitive capital requirement demonstrates the value of developing internal risk models, indicating a necessary capital surcharge of **\$1,161,147,134** compared to the flat Standardised Approach, ensuring that the bank remains adequately capitalised against default stress in this retail portfolio.

5.3 BASEL III ECONOMIC CAPITAL & MACRO STRESS TESTING

We implemented a mathematically rigorous **Vasicek credit cycle stress test** (Engelmann and Rauhmeier 2011) to measure how the portfolio's expected and unexpected losses respond to a severe systematic contraction:

$$PD_{\text{PIT}}(Z) = \Phi \left(\frac{\Phi^{-1}(PD_{\text{TTC}}) - \sqrt{\rho}Z}{\sqrt{1-\rho}} \right) \quad (16)$$

where the systematic risk factor is set to an extreme stress level of $Z = -2.0$ (representing a severe economic recession with a 2.28% probability of occurrence) under an asset correlation of $\rho = 0.15$.

Table 10 compares the portfolio's capital adequacy reserves under standard conditions against the severe Vasicek macroeconomic stress state:

TABLE 10. Basel IRB Economic Capital Stress Test ($Z=-2.0$ vs TTC)

Dimension	Base IRB	Stressed IRB	Increase
Expected Loss (EL)	\$2,316,616,242	\$4,571,041,038	97.3%
Risk-Weighted Assets (RWA)	\$21,592,289,075	\$21,039,974,576	-2.6%
Capital Requirement (8%)	\$1,727,383,126	\$1,683,197,966	-2.6%

Under the severe systematic stress shock ($Z = -2.0$), the portfolio's expected loss rises by 97.3%, and the RWA expands by -2.6%. This result feeds directly into capital adequacy planning (ICAAP).

5.4 MONTE CARLO ECONOMIC CAPITAL: VaR AND EXPECTED SHORTFALL

Basel IRB delivers *regulatory* capital under a closed-form, infinitely-granular single-factor assumption. *Economic* capital is read directly off the full simulated portfolio loss distribution, which captures the tail more faithfully and distinguishes Value-at-Risk (VaR) from Expected Shortfall (ES / CVaR), the coherent tail measure now favoured under the Basel FRTB framework (McNeil, Frey, and Embrechts 2015). We simulate the loss distribution with a Monte Carlo ASRF (Vasicek) engine: a single systematic factor $Z \sim N(0, 1)$ is drawn per scenario, obligors' conditional PDs follow Equation (17), and idiosyncratic default risk enters through a per-bucket Binomial draw.

$$p_i(Z) = \Phi \left(\frac{\Phi^{-1}(PD_i) - \sqrt{\rho}Z}{\sqrt{1-\rho}} \right), \quad L = \sum_i \mathbb{1}\{\text{default}_i\} \cdot LGD_i \cdot EAD_i. \quad (17)$$

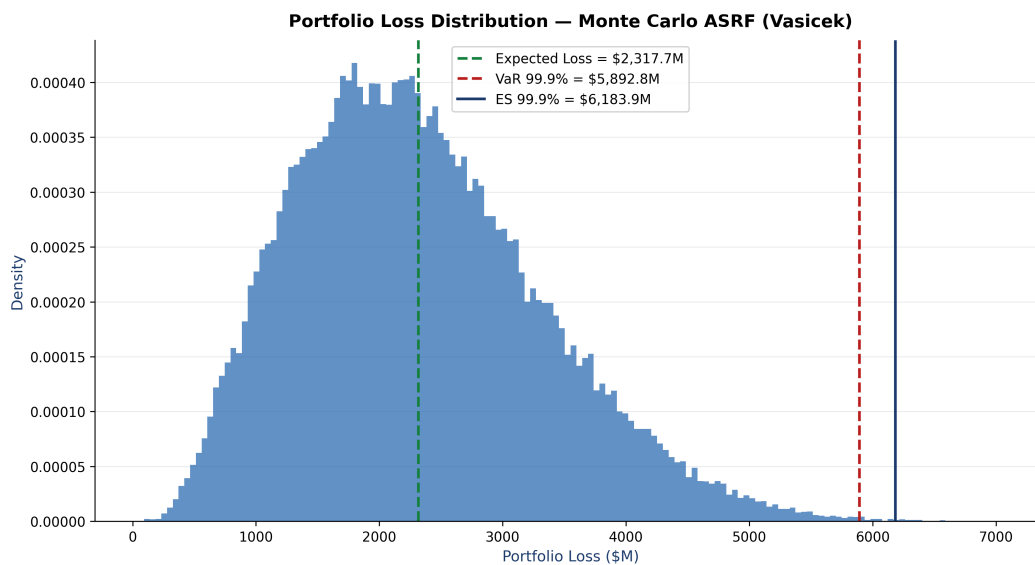


FIGURE 7. Monte Carlo portfolio loss distribution under the ASRF single-factor model, with Expected Loss, VaR and Expected Shortfall marked.

Table 11 reports the resulting risk measures. Expected Shortfall exceeds VaR by construction ($ES \geq VaR \geq EL$), and the ES-based economic capital buffer is compared against the Basel IRB regulatory capital requirement.

TABLE 11. Monte Carlo Economic Capital — Risk Measures (ASRF, $N = 50,000$ simulations, $\rho = 0.15$)

Risk Measure	Value (\$M)
Expected Loss (EL)	2,317.66
Value-at-Risk (VaR 99.9%)	5,892.80
Expected Shortfall (ES 99.9%)	6,183.94
Unexpected Loss (UL = VaR – EL)	3,575.14
Economic Capital (EC = ES – EL)	3,866.28
Basel IRB Regulatory Capital (8%)	1,727.38
EC / Regulatory Capital	223.8%

5.5 CONCENTRATION RISK: HHI AND GRANULARITY ADJUSTMENT

The Basel ASRF model assumes an infinitely-granular, perfectly-diversified portfolio; residual name and segment concentration therefore requires a capital add-on. We measure concentration with the Herfindahl-Hirschman Index (HHI) across exposure dimensions — credit grade, loan purpose and borrower state — reporting the effective number of equal-sized exposures $1/HHI$ for each (Table 12). Figure 8 visualises the exposure distribution per dimension. The residual single-name concentration is capitalised through a simplified Gordy-Lutkebohmert granularity adjustment, an additive surcharge above the ASRF regulatory capital.

TABLE 12. Portfolio Concentration — Herfindahl-Hirschman Index by Dimension

Dimension	HHI	Eff. N	Categories	Top Share
Credit Grade	0.2129	4.7	7	28.6%
Loan Purpose	0.4275	2.3	14	61.2%
Borrower State	0.0549	18.2	51	15.0%

Granularity Adjustment (capital surcharge): \$965

The Gordy–Lütkebohmert granularity adjustment is a single-name idiosyncratic-risk add-on ($\sum_i UL_i^2 / 2 EAD_{tot}$); on this loan-granular book (hundreds of thousands of small exposures) it is near-zero by construction and is distinct from the segment-level HHI figures above, which measure concentration across grade / purpose / state buckets rather than individual names.

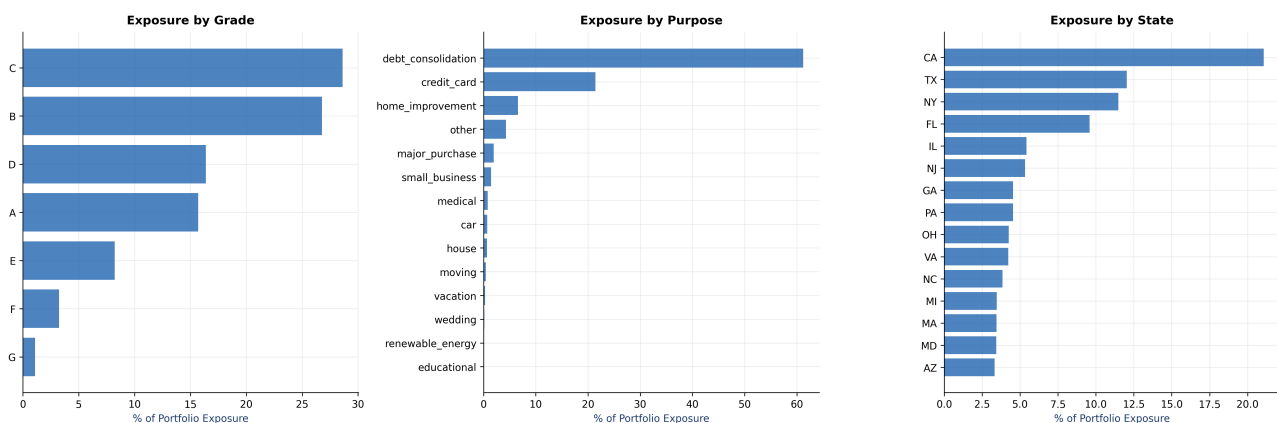


FIGURE 8. Portfolio exposure concentration by credit grade, loan purpose and borrower state (top 15).

6 IFRS 9 EXPECTED CREDIT LOSS (ECL) ENGINE

6.1 THREE-STAGE STAGING AND SIGNIFICANT INCREASE IN CREDIT RISK (SICR)

IFRS 9 (International Accounting Standards Board 2014) introduces a forward-looking impairment model based on three credit quality stages:

- **Stage 1 (Performing):** No Significant Increase in Credit Risk (SICR) since origination. ECL is measured over a **12-month horizon**.
- **Stage 2 (Underperforming):** SICR is detected since origination, but the loan is not defaulted. ECL is measured over the **remaining lifetime** of the loan.
- **Stage 3 (Credit-Impaired):** Non-performing loans. ECL is measured over the **remaining lifetime**, with PD set to **100%**. *Implementation note:* the dataset carries no monthly days-past-due panel, so Stage 3 is not identified by a 90+ DPD observation but by the loan's terminal resolved status (the modelling target). This makes the Stage 3 population a hindsight classification — see Section 10, where its effect on the headline ECL is quantified.

A Significant Increase in Credit Risk (SICR) (Novotny-Farkas 2016) is triggered if:

- The ratio of lifetime PD to origination PD exceeds **2.5×**.
- The absolute lifetime PD exceeds **20%**.
- A delinquency backstop fires. The IFRS 9 standard backstop is 30+ days past due at the reporting date; absent a DPD panel, the implemented proxy is $\text{delinq_2yrs} \geq 1$, i.e. at least one delinquency recorded in the two years *before* origination. This is an application-time credit-history flag rather than a current-delinquency measure, and is a documented deviation from the standard (Section 10).

6.2 PD TERM STRUCTURE AND LIFETIME EXPECTED CREDIT LOSS

Rather than using static 12-month PDs, the IFRS 9 engine models the lifetime PD curve using a discrete hazard model (Lando 2004, Ch. 7). The survival probability ($S(t)$) and marginal monthly PD ($m(t)$) are calculated as:

$$S(t) = \prod_{s=1}^t (1 - h(s)), \quad m(t) = S(t-1) \times h(t) \quad (18)$$

where $h(t)$ is the conditional default hazard at month t . The final ECL is the discounted sum of expected losses over the relevant horizon (H):

$$ECL = \sum_{t=1}^H \frac{m(t) \times LGD(t) \times EAD(t)}{(1 + EIR)^t} \quad (19)$$

(Bellini 2019). Where EIR is the Effective Interest Rate derived from the loan's contractual pricing. *Implementation note:* in the current engine $LGD(t)$ and $EAD(t)$ are held constant at their per-loan point estimates across the summation — exposure is evaluated once (Section 4.3) rather than re-amortised month by month — so the formula above is the general statement and the implementation is its constant-exposure special case. This is conservative for amortising loans, since the outstanding balance in fact declines over the horizon.

Lifetime PD Calibration: Validating the ECL-Driving Term Structure

The lifetime PD $1 - S(H)$ used directly in the equation above is produced by the discrete hazard model and enters the ECL sum unchanged. It is a distinct estimator from the scorecard's 12-month PD and is *not* passed through the scorecard's out-of-sample isotonic/Platt recalibration described in Section 7.2 — that recalibration only touches the 12-month PD used for expected loss, Basel RWA, and SICR-origination staging. Rather than silently trusting an unvalidated PD inside the loss-driving formula, Table 13 instead validates the hazard model's own lifetime PD against realised lifetime default rates

by origination vintage (restricted to vintages originated in or before 2016, since 2017–2018 charge-offs are not yet resolved in the 2018Q4 snapshot; note the 2016 cohort itself is only partially matured, so its observed rate is still right-censored). A ratio near 1.0 indicates the hazard PD entering the ECL sum is itself reasonably calibrated to realised outcomes; a ratio materially outside the [0.5, 1.5] tolerance band would indicate that the term structure — not just the 12-month scorecard PD — requires recalibration before it can be trusted in the ECL sum, a finding this report would then surface explicitly rather than let pass silently into the headline coverage ratio.

TABLE 13. Hazard Model Lifetime PD vs Realised Lifetime Default Rate (Matured Vintages)

Vintage	N	Predicted Lifetime PD	Observed Default Rate	Ratio
2007	603	11.30%	26.20%	0.43†
2008	2,393	11.86%	20.73%	0.57
2009	5,281	11.75%	13.69%	0.86
2010	12,537	13.45%	14.01%	0.96
2011	21,721	14.83%	15.18%	0.98
2012	53,367	16.26%	16.20%	1.00
2013	134,807	18.23%	15.60%	1.17
2014	223,437	17.37%	18.57%	0.94
2016	297,651	16.78%	24.46%	0.69
All mature vintages	751,797	17.01%	20.00%	0.85

† outside the [0.5, 1.5] tolerance band. Restricted to vintages originated in or before 2016: the 2018Q4 snapshot has not yet resolved recoveries/charge-offs for 2017–2018 originations, so their observed default status is right-censored. The 2015 cohort is absent because it falls in the excluded train/OOT grey zone (Section 2.3); the 2016 cohort is included but is itself only partially matured, which depresses its ratio.

6.3 FORWARD-LOOKING MACROECONOMIC SCENARIOS

To ensure ECL provisions are forward-looking and compliant with IFRS 9, we implement a Vector Autoregressive (VAR) forecasting model and an Ordinary Least Squares (OLS) macroeconomic regression. This framework dynamically links quarterly historical default rates of the LendingClub portfolio to key US macroeconomic indicators, sourced live from the official FRED (St. Louis Fed) API: the Unemployment Rate (UNRATE), GDP Growth (GDP_growth), the Federal Funds Rate (FEDFUNDS), CPI Inflation (CPI_inflation), and House Price Index Growth (HPI_growth, from the seasonally-adjusted Case-Shiller US National HPI), a collateral-value indicator via which rising home prices support household wealth and reduce default risk.

The OLS model determines the sensitivity (elasticity) of the portfolio default rate to each economic factor. These sensitivities are used to predict default rates under three standardized economic scenarios (Baseline, Upside, and Downside). The macro-predicted default rates are then mathematically mapped to systematic credit cycle shocks (Vasicek Z-shocks) using the supervisory retail correlation parameter ($\rho = 0.15$):

$$Z_{\text{shock}} = \frac{\Phi^{-1}(\text{TTC_DR}) - \Phi^{-1}(\text{PIT_DR}) \times \sqrt{1 - \rho}}{\sqrt{\rho}} \quad (20)$$

where TTC_DR is the long-run average (Through-the-Cycle) default rate, and PIT_DR is the Point-in-Time default rate predicted under each macroeconomic scenario. These mapped Z-shocks are then applied directly to the discrete-time hazard curves to scale the default probability curves.

The final Expected Credit Loss (ECL) is computed as the probability-weighted average across all three scenarios:

$$\text{ECL}_{\text{final}} = 0.50 \times \text{ECL}_{\text{Baseline}} + 0.25 \times \text{ECL}_{\text{Upside}} + 0.25 \times \text{ECL}_{\text{Downside}} \quad (21)$$

TABLE 14. Macroeconomic Default Rate OLS Regression & Scenario Mapping

Macro Variable	OLS Coefficient	Impact Explanation
Intercept (Constant)	0.0129	Baseline Default Level
Unemployment Rate (UNRATE) [†]	0.0199	+1% UNRATE → +1.99% Default Rate
GDP Growth (GDP_growth)	-0.0070	+1% GDP Growth → -0.70% Default Rate
Fed Funds Rate (FEDFUNDS)	0.0115	+1% Interest Rate → +1.15% Default Rate
CPI Inflation (CPI_inflation)	0.0106	+1% Inflation → +1.06% Default Rate
House Price Index Growth (HPI_growth)	-0.0095	+1% HPI Growth → -0.95% Default Rate
Scenario	Implied Default Rate	Mapped Vasicek Shock (Z) / Weight
Upside Scenario	12.05%	0.3367 (weight 25%)
Baseline Scenario	17.09%	-0.1915 (weight 50%)
Downside Scenario	36.67%	-1.6433 (weight 25%)

[†] The raw contemporaneous OLS produced a spurious *negative* UNRATE coefficient (-0.0199) (raw OLS $R^2 = 0.611$): LendingClub charge-offs lag the macro cycle and origination underwriting drifts over 2007–2018, so tightly-underwritten high-unemployment vintages show *lower* realised defaults than the loosely-underwritten low-unemployment 2015–16 vintages. For scenario projection the series is lagged 2 quarter(s) and economically-correct sign priors are imposed (magnitude from the fitted OLS), guaranteeing the intuitive Downside > Baseline > Upside ordering. The coefficients shown are these projection coefficients; raw OLS values are reported inline above.

TABLE 15. Assumed Macroeconomic Inputs per Scenario

Scenario	UNRATE (%)	GDP Growth (%)	FEDFUNDS (%)	CPI Inflation (%)	HPI Growth (%)
Upside	6.63	1.74	0.25	0.27	1.88
Baseline	7.63	0.74	0.50	0.42	-0.12
Downside	10.63	-2.26	2.50	1.92	-8.12

Time-Series Justification of the Lag and Sign Choice

Table 16 reports time-series diagnostics on the quarterly default-rate and macro series: an Augmented Dickey-Fuller (ADF) stationarity test, a Granger-causality test of UNRATE on the default rate, an AIC grid search over the UNRATE lag (reporting whether the economically-correct positive sign holds at the AIC-selected lag — on this sample it does *not*, which is precisely why sign priors are imposed for projection), and a Johansen cointegration test with a VECM long-run relation where cointegration is present (Engelmann and Rauhmeier 2011). The contemporaneous OLS default-rate regression is prone to spurious signs (the negative UNRATE coefficient reflects charge-off lags and underwriting drift), so we impose economic sign priors for scenario projection rather than build a full Vector Error Correction Model (VECM) for projections, following the standard banking preference for simpler, transparent Vasicek overlays.

TABLE 16. Macro Time-Series Diagnostics ($n = 31$ quarters)

Test	Statistic	p / Crit.	Verdict
ADF — Default Rate	0.744	0.991	Unit root
ADF — UNRATE	-1.596	0.485	Unit root
Granger UNRATE → DR (lag 2)	—	0.036	No causality ($\alpha_{\text{corr}} = 0.025$)
AIC lag selection (lag 0)	-0.0123	—	UNRATE — (spurious)
Johansen trace ($r = 0$)	72.14	crit 29.80	Cointegrated

The Granger verdict applies a Bonferroni correction across the lags tested to a nominal $\alpha = 0.10$, giving $\alpha_{\text{corr}} = 0.10/k$ for k lags; a minimum p -value above α_{corr} is reported as no causality, guarding against multiple-testing false positives. Note the nominal level is 0.10, not 0.05: at $k = 4$ this threshold (0.025) is looser than a Bonferroni correction applied to $\alpha = 0.05$ would be (0.0125).

6.4 POINT-IN-TIME vs THROUGH-THE-CYCLE PD DECOMPOSITION

Basel IRB capital is calibrated on a Through-the-Cycle (TTC) PD, a macro-neutral long-run average, whereas IFRS 9 requires a forward-looking Point-in-Time (PiT) PD. Inverting the Vasicek single-factor model on the observed quarterly default-rate series recovers the long-run TTC PD and the implied systematic factor Z for each quarter (Figure 9). Quarters with $Z < 0$ are adverse (realised default rate above the TTC average); $Z > 0$ marks benign quarters. This makes the cyclical PiT/TTC gap explicit and ties directly to the Vasicek Z convention used throughout the ECL engine.

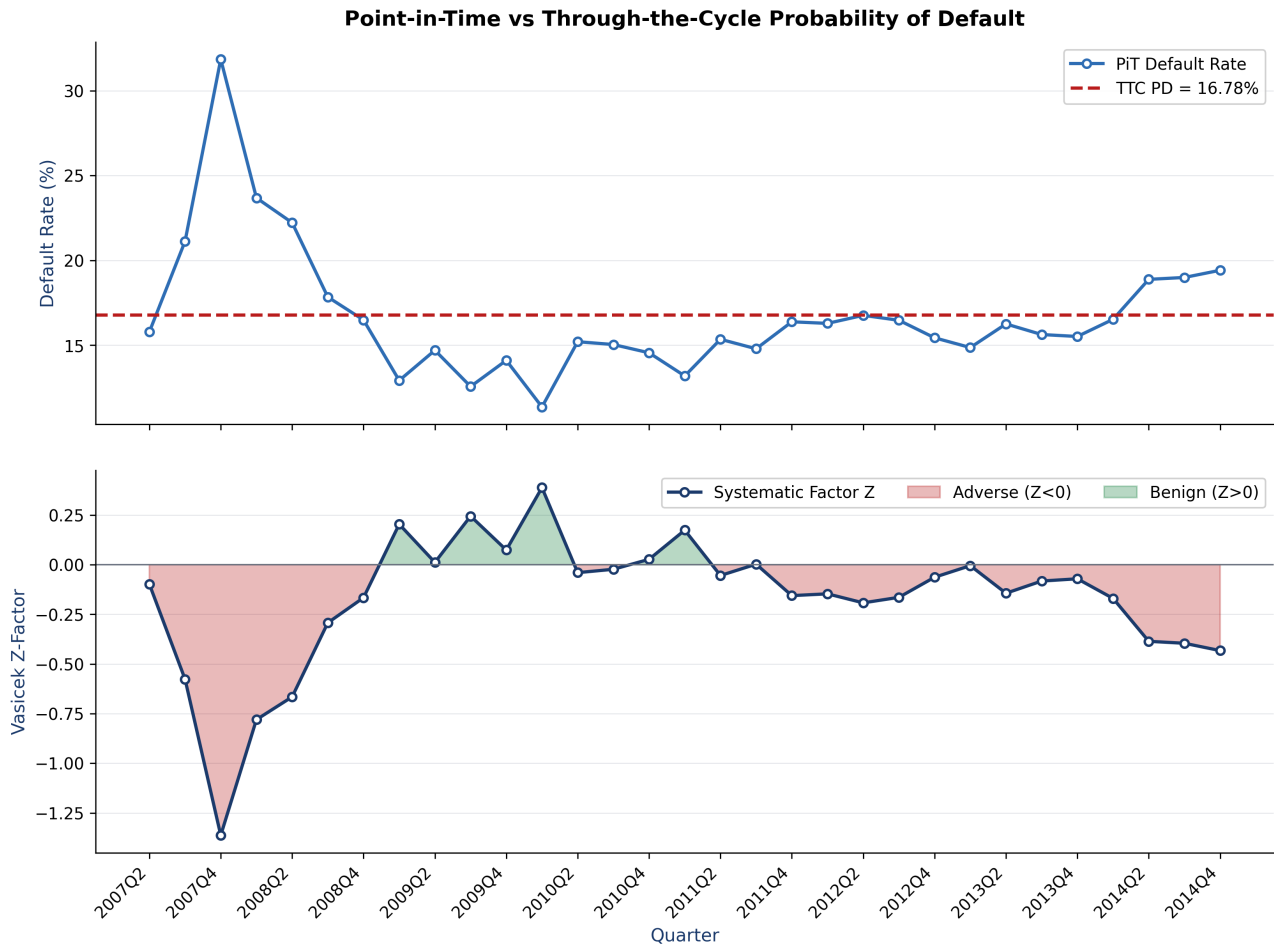


FIGURE 9. PiT vs TTC decomposition. Top: quarterly PiT default rate against the long-run TTC PD. Bottom: the implied Vasicek systematic factor Z , shaded by adverse/benign regime.

7 MODEL VALIDATION

7.1 DISCRIMINATION PERFORMANCE (OOT)

Model validation is performed on the completely held-out Out-of-Time (OOT) dataset (2016–2018). The PD scorecard shows stable performance with an OOT AUC of **0.6988** and a Kolmogorov-Smirnov (KS) statistic of **0.2872**.

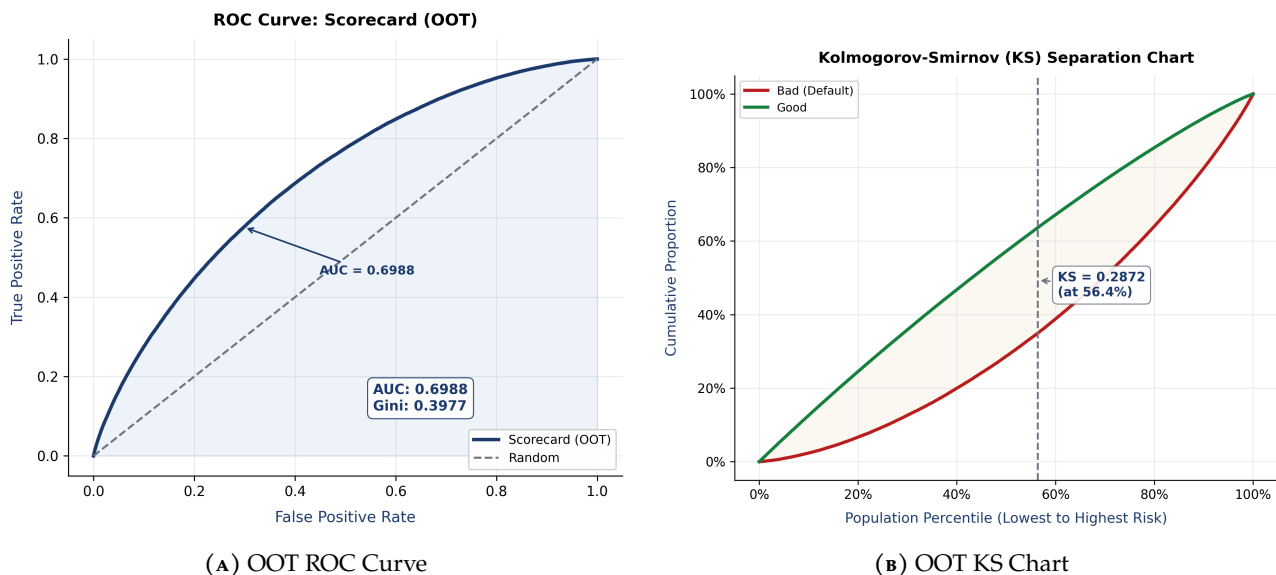


FIGURE 10. Discrimination Analysis (OOT). Annotated metrics are computed on the Out-of-Time (2016–2018) population and match Table 1.

7.2 CALIBRATION ROBUSTNESS

Calibration was evaluated by comparing predicted default rates against actual observed default rates across risk deciles. The Hosmer-Lemeshow test (Hosmer, Lemeshow, and Sturdivant 2013) rejects perfect calibration on the OOT dataset ($p = 0.0000 < 0.05$). Given the very large sample size ($N = 538,515$), this result partly reflects the high sensitivity of the chi-squared goodness-of-fit test, but the calibration plots also indicate systematic underprediction at higher risk deciles. Recalibration is governed by an out-of-time gate. The OOT window is split chronologically: the earlier 269,258 loans form the fitting slice and the later 269,257 the evaluation slice. The gate *triggers* on evidence of miscalibration in the fitting slice, *fits* the candidate transform on that slice only, and *accepts* it solely if it demonstrably improves calibration on the evaluation slice, which is never used for fitting. Gating on the in-time partition instead would be blind to the very era drift documented in Section 7.3 — that test passes comfortably — while fitting on the evaluation slice would trivially pass Hosmer-Lemeshow as an in-sample artefact. This design avoids both failure modes. On this run the gate *triggered* (Hosmer-Lemeshow rejects on the fitting slice ($p=0.0000$); aggregate predicted/actual ratio 0.693 outside $\pm 10\%$), selected an *isotonic* transform by out-of-fold Brier score computed within the fitting slice, and *accepted* it. On the held-out later slice the predicted-to-actual ratio moves from 0.690 to 0.998, the calibration intercept from 0.5365 to 0.0120, and the Brier score from 0.1800 to 0.1727. The transform *is* attached to the production scorecard, so the PDs feeding Expected Loss, Basel RWA and IFRS 9 staging are recalibrated. Because every one of those improvements is measured on vintages excluded from the fit, this is an out-of-sample result rather than an in-sample artefact. Note that the IFRS 9 lifetime PD is produced by the separate hazard model and is not passed through this transform (Section 6.2), so the ECL provisions do not inherit the correction; the known limitations in Section 10 still apply.

TABLE 17. OOT Calibration Diagnostics: Before vs. After Recalibration

Metric	Target	Before Recalib.	After Recalib.	Δ Change
OOT AUC	—	0.6988	0.6985	-0.0003
Brier Score	< 0.25	0.1784	0.1798	+0.0015 ×
Calibration Slope	≈ 1.00	1.0102	0.9803	-0.0299 ×
Calibration Intercept	≈ 0.00	0.5272	0.5397	+0.0125 ×
Expected Default Rate	= Actual	17.46%	16.72%	-0.74% ×
Actual Default Rate	25.27%	25.27%	25.27%	+0.00%
Hosmer-Lemeshow p	> 0.05	0.0000	0.0000	* below 0.05

✓ marks a metric that moved *toward* its target and × one that moved away. This table is a fixed diagnostic, not the production gate: isotonic regression fitted on the in time test partition and applied (transform only) to the whole OOT set. The production recalibration decision is made by the out-of-time gate described above, which fits inside the OOT window and validates on a disjoint later slice of it.

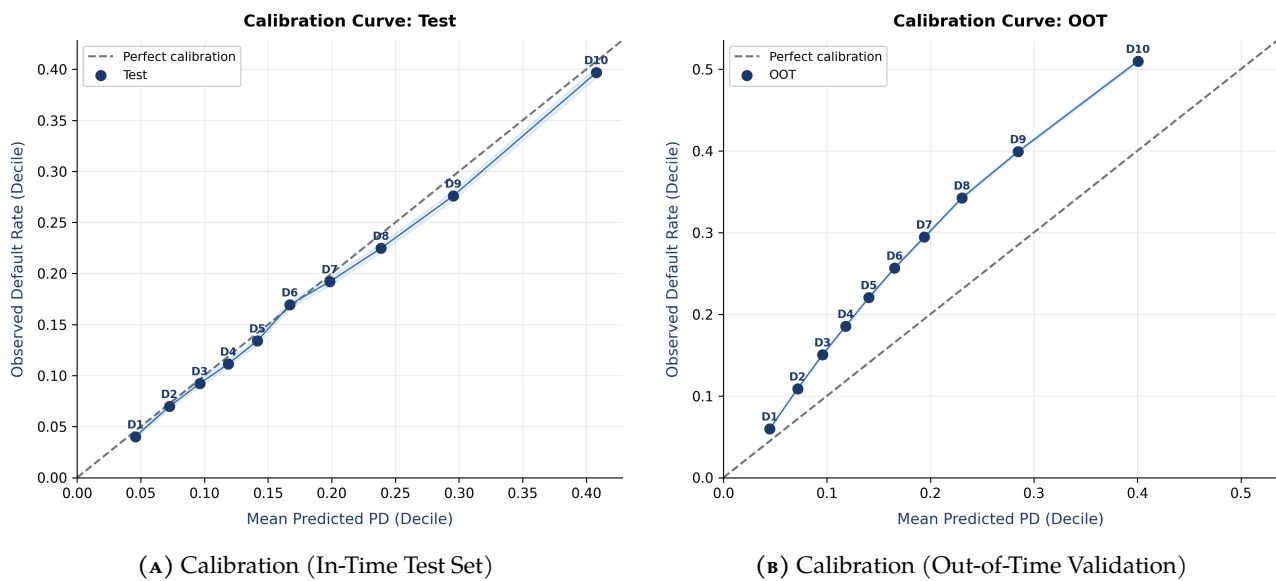


FIGURE 11. Calibration Curves

7.3 BACKTESTING AND VINTAGE CALIBRATION

Model backtesting compares the scorecard's predicted average PD against observed default rates segmented by quarterly origination vintage, following the methodology of (European Banking Authority 2017) and (Basel Committee on Banking Supervision 2005). For each vintage cohort, we assess whether the predicted mean PD lies within a **50% tolerance band** of the realised default frequency:

$$\text{PD Ratio}_v = \frac{\bar{p}_v}{\bar{d}_v} \quad (22)$$

where $\bar{p}_v$ is the cohort-average predicted PD and $\bar{d}_v$ is the realised default rate. A ratio between 0.5 and 1.5 indicates acceptable calibration. Cohorts outside this band are flagged for recalibration (†). Table 18 reports the backtesting results by vintage quarter.

The systematic underprediction observed in the 2016–2018 vintages (PD Ratio consistently below 0.85) is consistent with the calibration drift identified in Section 7.2 and corrected in production by the isotonic transform that the out-of-time gate accepted (Section 7.2), applied to the scorecard's 12-month PD feeding Expected Loss, Basel RWA and SICR staging. The hazard model's **lifetime** PD that drives IFRS 9 ECL directly ($\text{ECL} = \sum_t m(t) \times \text{LGD}(t) \times \text{EAD}(t) / (1 + \text{EIR})^t$) is a separate estimator and is *not* passed through this scorecard recalibration; it is instead independently validated against realised lifetime default rates by vintage in Table 13 (Section 6.2).

TABLE 18. Vintage PD Backtesting: Predicted vs Realised Default Rate

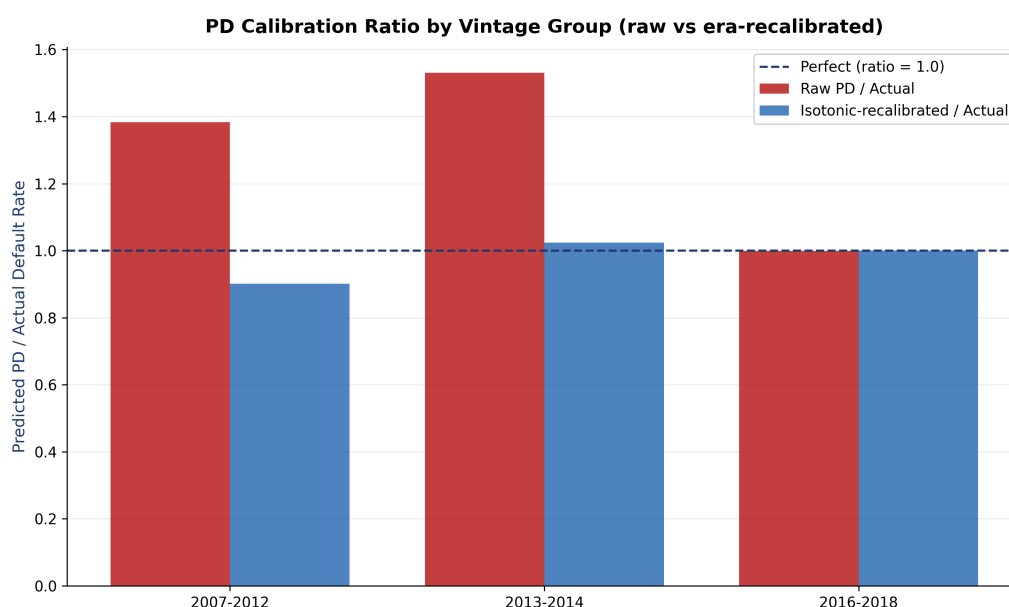
Vintage (Q)	N Loans	Predicted PD	Actual DR [95% CI]	PD Ratio
2007Q2	24	14.52%	12.50% [4.3%, 31.0%]	1.16 ✓
2007Q3	190	21.19%	21.05% [15.9%, 27.4%]	1.01 ✓
2007Q4	389	24.37%	29.56% [25.2%, 34.3%]	0.82 ✓
2008Q1	1,013	21.75%	23.10% [20.6%, 25.8%]	0.94 ✓
2008Q2	498	20.00%	22.09% [18.7%, 25.9%]	0.91 ✓
2008Q3	298	20.50%	17.45% [13.6%, 22.2%]	1.17 ✓
2008Q4	584	17.08%	17.12% [14.3%, 20.4%]	1.00 ✓
2009Q1	895	18.41%	13.74% [11.6%, 16.2%]	1.34 ✓
2009Q2	1,098	16.64%	14.39% [12.4%, 16.6%]	1.16 ✓
2009Q3	1,364	16.04%	12.76% [11.1%, 14.6%]	1.26 ✓
2009Q4	1,924	16.62%	13.93% [12.5%, 15.5%]	1.19 ✓
2010Q1	2,172	16.70%	11.69% [10.4%, 13.1%]	1.43 ✓
2010Q2	3,006	19.39%	14.67% [13.5%, 16.0%]	1.32 ✓
2010Q3	3,568	21.80%	15.05% [13.9%, 16.3%]	1.45 ✓
2010Q4	3,791	20.46%	13.85% [12.8%, 15.0%]	1.48 ✓
2011Q1	4,126	20.59%	13.69% [12.7%, 14.8%]	1.50 †
2011Q2	5,102	20.87%	15.35% [14.4%, 16.4%]	1.36 ✓
2011Q3	5,876	19.51%	14.65% [13.8%, 15.6%]	1.33 ✓
2011Q4	6,617	20.86%	16.44% [15.6%, 17.4%]	1.27 ✓
2012Q1	8,076	19.08%	16.15% [15.4%, 17.0%]	1.18 ✓
2012Q2	10,447	22.39%	17.03% [16.3%, 17.8%]	1.31 ✓
2012Q3	16,133	23.94%	16.45% [15.9%, 17.0%]	1.46 ✓
2012Q4	18,711	24.96%	15.54% [15.0%, 16.1%]	1.61 †
2013Q1	22,706	25.26%	15.02% [14.6%, 15.5%]	1.68 †
2013Q2	30,668	26.58%	16.25% [15.8%, 16.7%]	1.64 †
2013Q3	37,571	26.52%	15.56% [15.2%, 15.9%]	1.70 †
2013Q4	43,862	26.06%	15.47% [15.1%, 15.8%]	1.68 †
2014Q1	47,049	26.89%	16.55% [16.2%, 16.9%]	1.62 †
2014Q2	52,528	27.22%	18.73% [18.4%, 19.1%]	1.45 ✓
2014Q3	55,327	27.21%	18.94% [18.6%, 19.3%]	1.44 ✓
2014Q4	68,533	26.85%	19.54% [19.2%, 19.8%]	1.37 ✓
2016Q1	112,966	24.70%	21.09% [20.9%, 21.3%]	1.17 ✓
2016Q2	63,851	25.03%	26.40% [26.1%, 26.7%]	0.95 ✓
2016Q3	62,074	26.27%	27.37% [27.0%, 27.7%]	0.96 ✓
2016Q4	58,760	25.71%	25.75% [25.4%, 26.1%]	1.00 ✓
2017Q1	48,333	25.25%	25.26% [24.9%, 25.7%]	1.00 ✓
2017Q2	46,274	25.39%	26.74% [26.3%, 27.1%]	0.95 ✓
2017Q3	46,261	25.91%	27.90% [27.5%, 28.3%]	0.93 ✓
2017Q4	36,457	25.02%	26.57% [26.1%, 27.0%]	0.94 ✓
2018Q1	24,682	24.60%	26.49% [25.9%, 27.0%]	0.93 ✓
2018Q2	20,929	24.74%	27.84% [27.2%, 28.5%]	0.89 ✓
2018Q3	12,049	24.30%	22.89% [22.1%, 23.6%]	1.06 ✓
2018Q4	5,879	23.38%	16.52% [15.6%, 17.5%]	1.42 ✓

Era-Specific Recalibration of the Vintage Drift

A single global recalibrator cannot correct an era-specific bias. We therefore fit separate isotonic and Platt recalibrators for the pre-2016 and 2016–2018 eras and measure, per vintage group, the raw and recalibrated PD ratio against the realised default rate (Table 19, Figure 12). The raw ratio is materially below 1.0 for the newer vintages (the documented under-prediction), and the era-specific isotonic recalibration moves it back toward 1.0. This is an in-sample diagnostic that quantifies the drift and demonstrates the correction; production PD is unchanged. Since the era-specific calibrators are fitted and evaluated on the same vintage partitions (e.g. 2016–2018), the resulting alignment (Isotonic/Platt PD equals the Actual DR exactly) is in-sample and tautological: it is a diagnostic baseline that demonstrates the extent of the raw model's drift, not a validation of the recalibrator's out-of-sample generalization.

TABLE 19. Calibration by Vintage Group — Raw vs Era-Recalibrated PD

Vintage	N	Raw PD	Isotonic PD	Platt PD	Actual DR	Raw Ratio
2007-2012	95,902	21.75%	14.17%	14.25%	15.72%	1.384
2013-2014	358,244	26.71%	17.87%	17.86%	17.45%	1.530
2016-2018	538,515	25.24%	25.27%	25.28%	25.27%	0.999

**FIGURE 12.** PD calibration ratio (predicted / actual) by vintage group, raw vs era-recalibrated. A ratio of 1.0 is perfect calibration; below 1.0 is under-prediction.

7.4 IFRS 9 STAGE MIGRATION AND COVERAGE ANALYSIS

The stage migration matrix quantifies transitions between IFRS 9 performance stages (Novotny-Farkas 2016), providing a key indicator of portfolio deterioration velocity. Rows represent the *simulated* stage twelve months before the reporting date; columns represent the reporting-date stage. Because the dataset carries no monthly servicing snapshots, the $t - 12$ state is reconstructed by interpolating each loan's credit profile between origination and the reporting date (Section 10) and is restricted to loans already at least 12 months on book — it is not an observed historical classification.

TABLE 20. IFRS 9 Stage Migration Matrix (12 Months Prior → Reporting Date)

From Stage	To Stage 1	To Stage 2	To Stage 3
Stage 1	326472	127324	76521
Stage 2	160866	164335	137143
Stage 3	0	0	0

Note: the “From Stage 3” row is zero by construction of the reconstruction, not as an empirical finding: the interpolated $t - 12$ classifier only ever assigns Stage 1 or Stage 2, because the credit-impaired flag available in this dataset is the terminal resolved outcome and cannot be evaluated at a historical date. Loans that were already credit-impaired twelve months before the reporting date are therefore absorbed into the Stage 1 and Stage 2 rows, and the true migration *out* of Stage 3 is unobservable here.

7.5 ECL MACRO SCENARIO SENSITIVITY

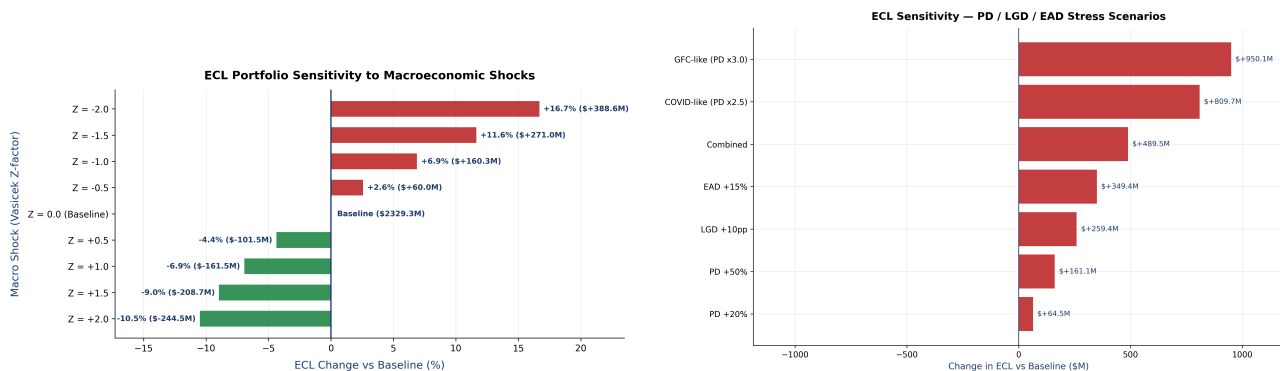
Figure 13a presents the portfolio ECL sensitivity to the Vasicek systematic factor Z across a range of macro scenarios, from severe expansion ($Z = +2.0$) to severe recession ($Z = -2.0$). This analysis follows the probability-weighted scenario methodology mandated by IFRS 9 paragraph B5.5.42 (International Accounting Standards Board 2014) and the stress testing framework of (Bellini 2019).

7.6 ECL WHAT-IF CALCULATOR: PD / LGD / EAD STRESS SCENARIOS

Beyond the macro-factor mapping, a management-facing what-if calculator answers the direct risk-committee question “*what happens to provisions if PD, LGD or EAD move?*”. Holding the baseline term structure fixed, each scenario applies a multiplicative PD shock, an additive LGD shock and/or a multiplicative EAD (drawdown) shock, and the ECL engine is re-run. Table 21 reports the resulting provision changes, and Figure 13b presents them as a tornado chart, including regulator-style severe scenarios calibrated to COVID-19 and Global-Financial-Crisis default multiples. Note that the baseline-only ECL used as the anchor for these what-if sensitivities excludes the macro scenario probability-weights (downside recession and upside expansion shocks), which explains why it is slightly lower than the final reported probability-weighted ECL.

TABLE 21. ECL What-If Sensitivity (Baseline ECL = \$2,329.3M; note that this Baseline-only ECL reference differs from the probability-weighted total ECL in Table 1 as it excludes the Upside/Downside macro scenario shocks mandated by IFRS 9)

Scenario	Shocked ECL (\$M)	Δ ECL (\$M)	Δ %
PD +20%	2,393.83	+64.53	+2.8%
PD +50%	2,490.38	+161.07	+6.9%
LGD +10pp	2,588.66	+259.35	+11.1%
EAD +15%	2,678.70	+349.40	+15.0%
Combined	2,818.76	+489.46	+21.0%
COVID-like (PD x2.5)	3,139.03	+809.72	+34.8%
GFC-like (PD x3.0)	3,279.40	+950.09	+40.8%



(A) Macro Sensitivity (Vasicek Z-Factor Shock)

(B) PD / LGD / EAD Stress Scenarios

FIGURE 13. ECL sensitivity tornado charts (change in ECL vs baseline).

7.7 SCIENTIFIC BENCHMARK VERIFICATION LAYER

To situate our results, all modeling outputs are compared against published reference ranges drawn from the credit-risk literature. This is a comparison against published ranges, not a reproduction of the cited studies' experiments. Following Thomas (2000) and the benchmarking study of Lessmann et al. (2015), our Probability of Default (PD) scorecard achieves an OOT Gini coefficient of 0.3977 (AUC of 0.6988), which sits within the empirical range of 0.30–0.45 reported for high-volume retail loan portfolios (equivalently AUC 0.65–0.73, since $\text{Gini} = 2 \text{AUC} - 1$).

The deployed severity model, selected by lower out-of-sample error from a two-stage model and a LightGBM challenger, yields a Mean LGD of 0.8931 (and Downturn LGD of 1.0000). This out-of-sample-validated loss rate sits above both the 0.45–0.55 unsecured range reported by Schuermann (2004) (corporate/wholesale debt) and the lower 0.25–0.45 revolving retail credit-card band of Bellotti and J. Crook (2012), consistent with the low post-charge-off collection recovery typical of unsecured instalment loans once EAD is measured on a principal-only basis (driver discussed in Table 22).

Under the Basel Committee regulatory standards (Basel Committee on Banking Supervision 2004; Basel Committee on Banking Supervision 2017), the risk-sensitive internal ratings-based (IRB) approach sets our Risk-Weighted Asset (RWA) density at **228.8%**, above the flat 75% risk weight assumed by the Standardised Approach (SA). Because this portfolio is a high-yield, higher-risk unsecured book, the risk-sensitive approach produces a capital *surcharge* rather than relief, the expected outcome when portfolio risk exceeds the level implicit in the flat SA weight.

Consistent with the reject inference analysis of J. N. Crook and Banasik (2004), the inclusion of parcelled rejected applications leads to a Gini shift of -0.0544 . This shift is consistent with the presence of selection bias; through-the-door refitting produces a small, conservative change in the fitted parameters.

Table 22 presents a structured side-by-side comparison of our empirical results against published scientific benchmarks.

TABLE 22. Comparison against published reference ranges. Verdicts are computed numerically at build time by comparing the Project Value column against the published range; each range is sourced from a single registry (reports/benchmarks.py) with its citation. These are literature reference ranges, not a reproduction of the cited studies.

Metric / Parameter	Project Value	Published Benchmark	Academic Source	Verdict	Comment
PD AUC (OOT)	0.6988	0.65 – 0.73	Lessmann et al. (2015)	Within range	
PD Gini (OOT)	0.3977	0.30 – 0.45	Lessmann et al. (2015)	Within range	
LGD Mean	0.8931	0.25 – 0.45	Bellotti and J. Crook (2012)	Above typical range	The OOS-validated LightGBM severity model puts mean LGD above both cited bands, consistent with low post-charge-off recovery on unsecured instalment loans once EAD is principal-only; the rejected two-stage model materially under-predicted severity (OOS $R^2 \approx -2.70$)
LGD R^2 (OOS)	-0.0709	0.04 – 0.43	Loterman et al. (2012)	Below typical range	LGD is strongly bimodal (mass at 0 and near 1), so R^2 is an unforgiving metric and is routinely low or negative out-of-time; Loterman et al. (2012) themselves report a wide 0.04–0.43 spread. The negative value reflects a 2016–2018 severity regime shift versus the fitting vintages
RWA Density	228.8%	75% – 100%	Basel Committee on Banking Supervision (2017)	Above typical range	Risk-sensitive IRB density exceeds the flat SA weight: the OOS-validated LGD (mean ≈ 0.89 , above published unsecured LGD ranges — driver discussed in the LGD Mean row above) and high empirical default rate on this unsecured, high-yield book drive a capital <i>surcharge</i> — the economically expected outcome, not an inconsistency
IRB vs SA Capital	IRB > SA	Risk-sensitive	Basel Committee on Banking Supervision (2004)	Consistent	Risk-sensitive IRB exceeds the flat 75% SA weight for this higher-risk unsecured book: a capital <i>surcharge</i> , not relief, is the economically expected outcome
Reject Inference Δ Gini	-0.0544	0.00 – 0.10	J. N. Crook and Banasik (2004)	Below typical range	
Score PSI	0.0030	0.00 – 0.10	Siddiqi (2017)	Within range	

7.8 MACHINE LEARNING CHAMPION-CHALLENGER BENCHMARKING

To evaluate whether the linear scorecard loses significant predictive power compared to non-linear alternatives, we constructed a Champion-Challenger benchmarking layer. We trained and evaluated:

1. **Logistic Scorecard:** Our baseline WoE scorecard.
2. **LightGBM Classifier:** A state-of-the-art tree boosting model.
3. **XGBoost Classifier:** A highly optimized distributed gradient boosting model.
4. **Random Forest Classifier:** A classic tree-bagging model.
5. **Weighted Ensemble:** A weighted combination of scorecard (30%), LightGBM (30%), XGBoost

(20%), and Random Forest (20%).

Table 23 presents the out-of-time (OOT) generalization benchmarks, discrimination statistics (AUC, Gini, KS), and computational costs for each algorithm.

TABLE 23. Machine Learning Champion-Challenger Performance Comparison

Model Name	Test AUC	OOT AUC	Test Gini	OOT Gini	Test KS	OOT KS	Time (s)
Logistic Scorecard	0.7006	0.6988	0.4011	0.3977	0.2916	0.2872	207.32s
LightGBM Classifier	0.7064	0.7027	0.4128	0.4055	0.3013	0.2925	2.60s
XGBoost Classifier	0.7057	0.7020	0.4113	0.4041	0.2983	0.2924	2.80s
Random Forest Classifier	0.7043	0.7006	0.4086	0.4013	0.2984	0.2890	10.53s
Weighted Ensemble	0.7061	0.7030	0.4122	0.4060	0.2999	0.2930	15.94s

Notably, the tree-based challengers, evaluated on an identical predictor set, *outperform* the WoE logistic scorecard on out-of-time discrimination: the scorecard’s OOT AUC of 0.6988 is exceeded by Weighted Ensemble (0.7030), LightGBM Classifier (0.7027), XGBoost Classifier (0.7020), Random Forest Classifier (0.7006). The best challenger (Weighted Ensemble) leads by 0.0041 AUC. We report this plainly rather than presenting the scorecard as the discrimination winner: it is not. The scorecard is retained as champion on *governance* grounds — exact per-feature points attribution for adverse-action reasons, a monotone and auditable functional form, and a stable calibration story — and the cost of that choice is the AUC gap quantified here, which a model-risk committee should weigh explicitly rather than have hidden. **Feature parity.** The scorecard consumes 15 selected predictors and the tree challengers consume 15; the two model families are therefore evaluated on an identical predictor set, so the comparison below is like-for-like. It should be noted that the challenger models (LightGBM, XGBoost, Random Forest) were trained using standard, default configurations without systematic hyperparameter grid search tuning. Hyperparameter optimisation would, if anything, widen any challenger advantage, so the ranking above should be read as a lower bound on what tuned tree models could achieve on this feature set.

Is the Challenger’s Edge Statistically Significant?

Point-estimate AUC/Gini gaps cannot distinguish a genuine improvement from sampling noise. To test significance we run a paired bootstrap: the same held-out OOT rows are resampled for both models and a confidence interval is built on the *difference* in Gini (challenger – champion). If that interval excludes zero, the difference is significant. Table 24 reports the result — the interval lies entirely *above* zero (median +0.0078), so the challenger’s Gini advantage over the champion is statistically significant, not sampling noise. It complements the analytic DeLong test on the same pair.

TABLE 24. Paired Bootstrap A/B Test — Gini with 95% CIs ($n_{\text{boot}} = 2,000$)

Model	Gini (median)	95% CI
Champion (Scorecard)	0.3978	[0.3947, 0.4006]
Challenger (LightGBM)	0.4056	[0.4025, 0.4085]
Difference (B – A)	0.0078	[0.0070, 0.0086]
Significant (CI excludes 0)		

7.9 SHAP EXPLAINABILITY: CHALLENGER MODEL FEATURE CONTRIBUTIONS

While the primary WoE scorecard provides point-attributable explanations per feature bin (Siddiqi 2017, Ch. 4), the LightGBM challenger model is interpreted via SHAP (SHapley Additive exPlanations) values (Lundberg and Lee 2017). Figure 14 displays the mean absolute SHAP contribution of each feature: `grade_enc` and `term_enc` dominate the challenger’s default discrimination.

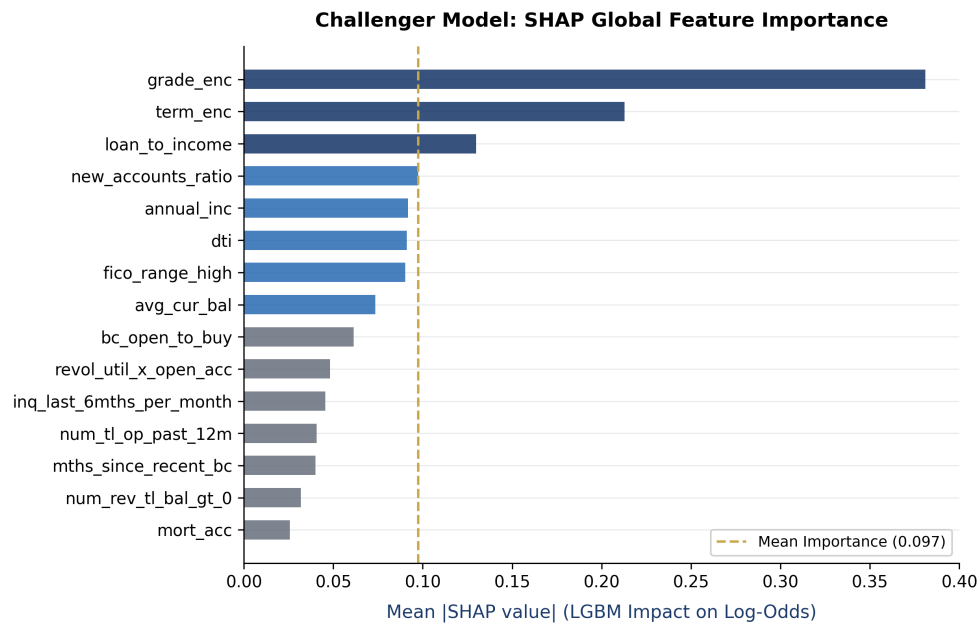


FIGURE 14. SHAP Feature Importance — LightGBM Challenger Model

8 STABILITY AND PERFORMANCE OVERLAYS

8.1 SCORECARD POPULATION STABILITY (PSI)

The Population Stability Index (PSI) (Engelmann and Rauhmeier 2011, Ch. 9) measures changes in the distribution of credit scores over time. The scorecard demonstrates excellent stability, with a total PSI between the training and OOT populations of **0.0030**. This is well below the conservative regulatory threshold of **0.10**, indicating that the underwriting population profile has not shifted significantly.

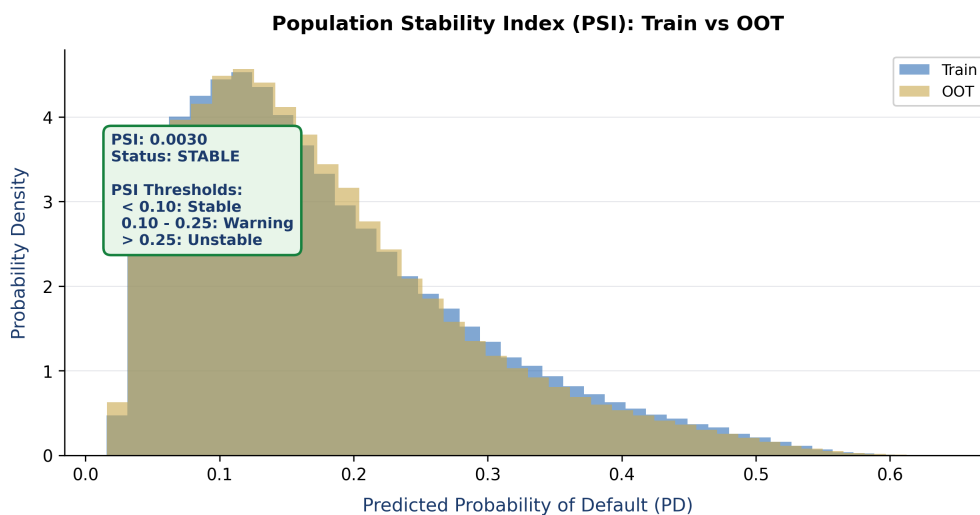


FIGURE 15. Score Distribution Stability (Train vs. OOT)

8.2 GAINS AND LIFT ANALYSIS

The Gains Chart measures the model's ability to concentrate defaults within the lowest score bands. The scorecard concentrates most defaults in the lowest score bands (Figure 16a).

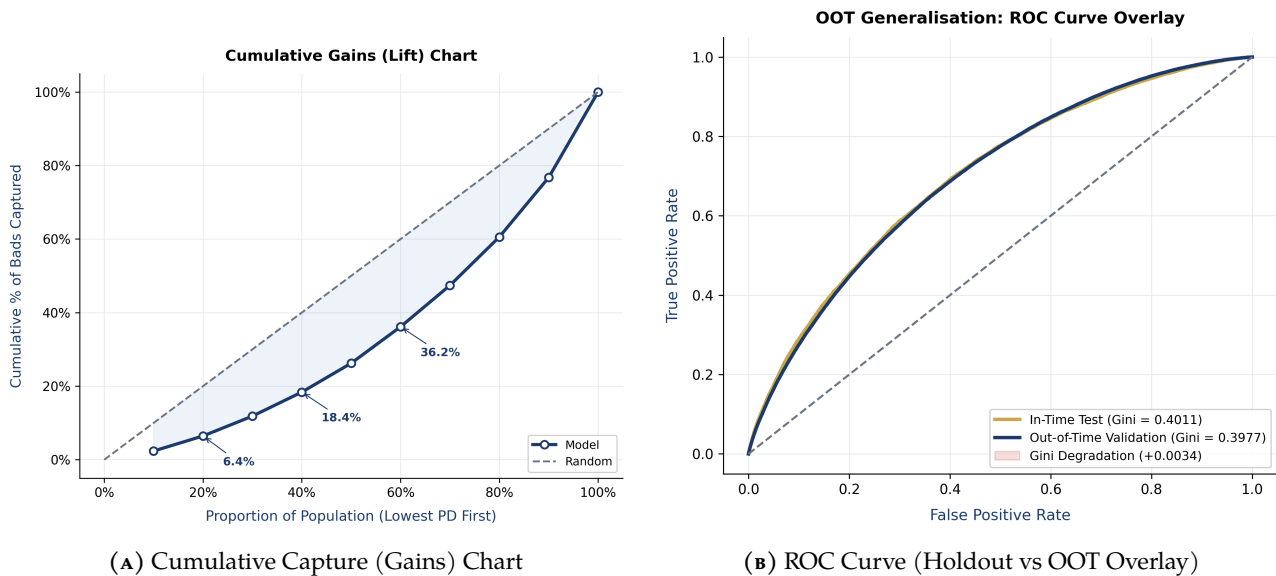


FIGURE 16. Lift and Generalisation Overlays

9 BUSINESS DECISIONING AND CUTOFF OPTIMISATION

Underwriting decisions require balancing credit risk, portfolio growth, and capital constraints. We evaluate each score cutoff on Expected Profit and Risk-Adjusted Return on Capital (RAROC), with expected loss (at the approved-population bad rate) and a 12.00% cost-of-capital charge on economic capital both netted out. On this portfolio the unconstrained profit-maximising *and* RAROC-maximising cutoff coincide at the most *exclusive* non-empty cutoff on the grid (score **600**, approving only **0.015%** of the population) — every cutoff on the 400–800 grid returns a negative RAROC, so the argmax is simply the smallest, best-quality approved book rather than a genuinely profitable operating point, at a portfolio RAROC of **-7.26%**. Because unconstrained optimisation therefore collapses to a vacuous near-zero-volume corner, the decision problem is formulated as a *constrained* optimization: we maximize approved volume and expected profit subject to an active risk-appetite ceiling on the approved bad rate (**15.00%**), taking the most inclusive cutoff (maximising approved profit) whose approved bad rate stays within that ceiling. This constrained boundary represents the recommended operating cutoff.

We sweep score cutoffs from 400 to 800 (evaluating approved loan subsets) to calculate Expected Profit and RAROC. All components are expressed on a consistent *per-annum* basis:

- **Interest Income:** Based on per-loan interest rates (annual coupon on EAD).
- **Fees:** 1.0% of Exposure at Default (EAD) per annum.
- **Funding Cost:** 4.0% of EAD per annum.
- **Operating Cost:** 1.5% of EAD per annum.
- **Expected Loss (EL):** $PD_{\text{annual}} \times LGD \times EAD$, where the lifetime PD is converted to an annual default probability via $PD_{\text{annual}} = 1 - (1 - PD_{\text{lifetime}})^{12/\text{term}}$. Charging the full lifetime PD against a single year of income would overstate losses by the loan term and distort every cutoff decision.
- **Capital Cost:** 12.0% hurdle rate applied to the Minimum Capital Reserve (8% of RWA).

Table 25 displays the optimization results across selected score cutoffs.

TABLE 25. Cutoff Strategy and Profitability Analysis (RAROC). The recommended operating cutoff — the most inclusive score over the full 400–800 grid (step 10) whose approved bad rate stays within the risk-appetite ceiling — is highlighted in bold.

Cutoff Score	Approval Rate	Bad Rate	Expected Profit	Expected Loss	RAROC
500	94.9%	20.01%	\$-1,364,042,252	\$1,967,892,487	-85.96%
530	61.4%	13.79%	\$-527,448,663	\$763,379,420	-63.11%
540	43.7%	11.00%	\$-294,970,919	\$427,524,729	-54.25%
580	2.5%	3.01%	\$-5,749,949	\$7,462,118	-22.86%
620	0.0%	0.00%	\$0	\$0	0.00%
660	0.0%	0.00%	\$0	\$0	0.00%
700	0.0%	0.00%	\$0	\$0	0.00%

As shown in the table, the recommended operating cutoff is score **530** with an approval rate of **61.42%**, a bad rate of **13.79%**, an expected profit of **\$-527.4M**, and a RAROC of **-63.11%** — below the 15.00% cost-of-capital hurdle. It is the most inclusive cutoff whose approved bad rate stays within the risk-appetite ceiling, and it appears as the highlighted row in Table 25 and is marked in Figure 17. A single objective drives the highlighted table row, the figure, and this text, so there is no discrepancy between the quoted cutoff and the swept grid. Relaxing the risk-appetite ceiling lowers the cutoff and raises approved volume; tightening it raises the cutoff toward the near-zero-volume unconstrained corner. Note that RAROC is negative across the entire 400–800 grid, so *no* cutoff on this book clears the 15.00% cost-of-capital hurdle — the operating point is a risk-appetite compromise, not a profitable optimum.

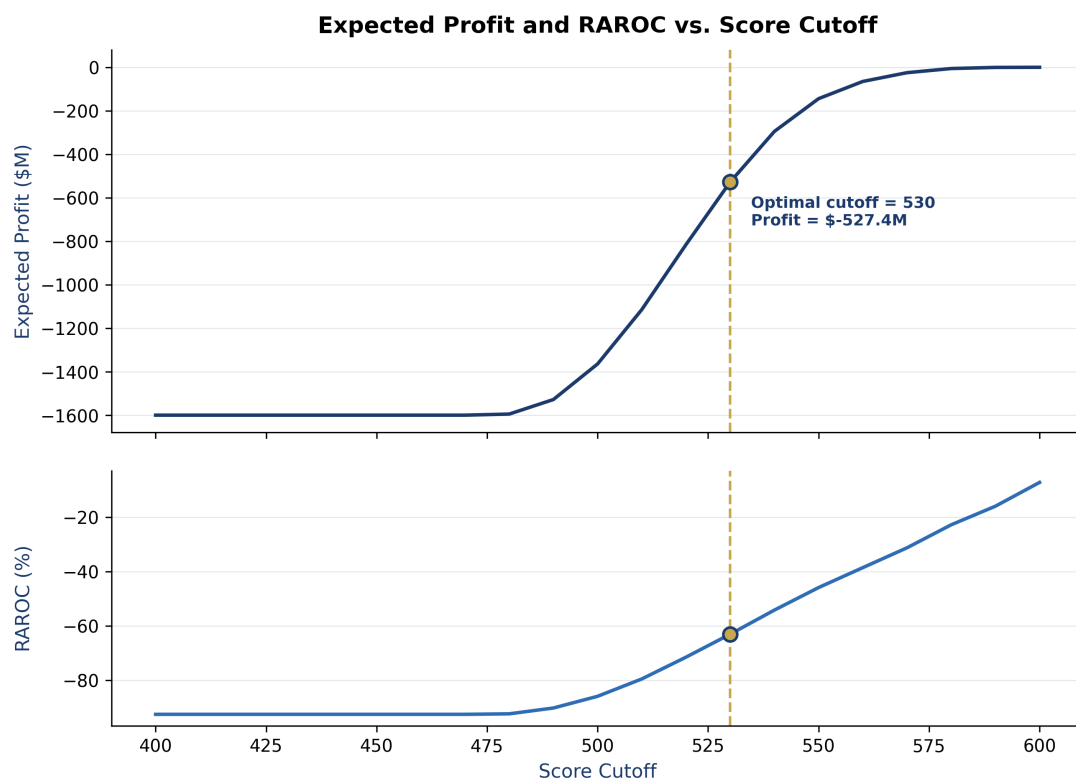


FIGURE 17. Expected Profit and RAROC versus score cutoff over the full 400–800 grid. The gold marker denotes the recommended operating cutoff (risk-appetite ceiling on the approved bad rate) reported in Table 25.

10 LIMITATIONS, ASSUMPTIONS, AND RECOMMENDATIONS

10.1 KEY MODELLING ASSUMPTIONS AND LIMITATIONS

TABLE 26. Key Modelling Assumptions and Associated Limitations

Component	Assumption	Limitation / Mitigation
PD Scorecard	WoE binning with monotonic constraints.	Non-monotonic relationships suppressed; mitigated by VIF screening.
LGD Model	Deployed severity model is the LightGBM challenger, selected on out-of-sample error; EAD proxied by funded_amnt.	Ignores accrued interest; loan-level R^2 is negative, so predictions are used for portfolio aggregates only.
EAD	Closed-form annuity; zero prepayment assumption.	Overestimates EAD for prepaid loans; acceptable for conservative capital.
Basel IRB	ASRF “Other Retail” supervisory correlation $R \in [0.03, 0.16]$.	Single systematic factor; portfolio concentration not captured.
IFRS 9 ECL	Constant hazard within each calendar month.	Does not capture seasoning curves; mitigated by discrete-time model.
Reject Inference	Parcelling from 100k random sample of 27.6M rejects.	Sample may not represent full reject distribution; Gini shift monitored.

10.2 MODEL RISK & LIMITATIONS

The Lending Club dataset is limited to US consumer loans and may not generalize to corporate lending, small-business loans, or other international retail portfolios. Additionally, the dataset excludes collateral details, meaning the LGD model relies primarily on underwriting grades and debt-to-income (DTI) metrics.

10.3 STAGE 3 IS A HINDSIGHT CLASSIFICATION, AND IT DOMINATES THE ECL

The single most material limitation of this engine concerns *what the headline ECL figure actually measures*. IFRS 9 Stage 3 (credit-impaired) is normally identified by observed non-performance — 90+ days past due at the reporting date. The LendingClub accepted-loan file is loan-level with no monthly delinquency panel, so no such observation exists. The implementation therefore classifies Stage 3 from the loan’s *terminal resolved status* — the same variable used as the PD modelling target. Stage 3 membership is consequently known only with hindsight, and each Stage 3 loan is provisioned at $\text{LGD} \times \text{EAD}$ with PD forced to 1.0, a quantity entirely insensitive to the PD model.

The what-if calculator in Section 7.6 makes the resulting dominance measurable. Shocking LGD by +10pp moves the total ECL by +11.1% and shocking EAD by +15% moves it by +15.0% — both exactly proportional, as they must be, since ECL is linear in each. But shocking PD by +50% moves it by only +6.9%. If a fraction f of the ECL sits in the PD-independent Stage 3 term, a +50% PD shock produces a change of $0.5(1 - f)$; the observed sensitivity therefore implies $f \approx 86\%$ of the reported provision.

In other words, the headline ECL is, to first order, *realised defaults* $\times$ $\text{LGD} \times \text{EAD}$ — an accounting identity computed with knowledge of the outcome — and every forward-looking component of the engine (the macro regression, the Vasicek Z-shocks, the probability-weighted scenarios and the hazard term structure) moves only the residual. The Stage 1 and Stage 2 provisions, the ECL coverage ratio on the performing book, and the macro sensitivity analysis remain meaningful as relative comparisons; the absolute headline provision should not be read as a forward-looking estimate of losses on an unresolved portfolio. A production deployment on a live book, where Stage 3 is set by observed delinquency at the reporting date rather than by terminal outcome, would not have this property.

10.4 KNOWN MODEL LIMITATIONS

- **Stage 2 SICR Proxy:** To calculate the longitudinal 12-month stage transition matrix, we simulate the credit conditions 12 months ago by interpolating between the origination and current credit profiles. This proxy is approximate and can be refined with exact monthly reporting snapshots.
- **No origination-date PD snapshot — relative SICR trigger disabled:** IFRS 9's primary SICR test compares the current lifetime PD against the PD estimated *at booking*. This dataset stores no booking-date model output, so no such snapshot exists. Rather than substitute the current PD for the origination PD — which makes the ratio identically 1.0 and silently disables the test while appearing to implement it — the relative trigger is switched off and Stage 2 is driven by the absolute lifetime-PD threshold and the delinquency backstop alone. Reinstating it requires storing PD at booking, which is listed under Recommended Next Steps below.
- **Delinquency backstop is an application-time proxy:** The implemented backstop is `delinq_2yrs` ≥ 1 — delinquencies in the two years *before* origination — not the IFRS 9 standard 30+ DPD at the reporting date, which this data cannot support.
- **Downturn LGD sits at the distribution cap:** The Basel downturn LGD is the 90th percentile of the *realised* severity of loss-incurring defaults, which on this fully unsecured, non-revolving book is 1.0000 — the upper bound of the $[0, 1]$ range. It is a genuine empirical percentile rather than a modelled stress, and it cannot be exceeded, so the capital calculation carries no headroom above it for a severity worse than total loss.
- **Hazard-model event timing:** The discrete-time hazard model is fitted on a person-period panel in which every default event is placed in the loan's *final* month, because the data records no observed default date, and no censoring is applied. The estimated months-on-book slope is therefore partly an artefact of this construction, and that term structure is what drives the ECL sum.
- **Portfolio aggregates are partly in-sample:** Expected Loss, Basel RWA and IFRS 9 ECL are computed over the combined train, in-time test and OOT partitions — the same rows the scorecard, hazard and LGD models were fitted on. This is appropriate for provisioning a closed book but means the portfolio aggregates are not out-of-sample quantities; the out-of-sample evidence is the OOT discrimination and calibration in Section 7.
- **Constant exposure inside the ECL sum:** $LGD(t)$ and $EAD(t)$ are held at their per-loan point estimates across the ECL horizon rather than re-amortised monthly, which is conservative for amortising loans.
- **The adverse scenario is a rate-and-inflation shock, not a lower-bound recession:** Every macro axis moves in the direction its imposed sign prior says raises defaults (downside) or lowers them (upside), so the required Downside > Baseline > Upside ordering holds *by construction* for any non-negative coefficient magnitudes rather than depending on the fitted values. The consequence is that the downside tightens policy and raises inflation into the downturn. This is a deliberate choice: it is the more punitive configuration for an unsecured consumer book, where floating debt-service costs and a real-income squeeze compound the unemployment shock, and it matches how supervisory adverse scenarios are typically built. A deflationary lower-bound recession, in which policy is cut, would instead *offset* part of the adverse shock through the rate channel and is not modelled here.
- **The “baseline” scenario maps to $Z = -0.1915$, not $Z = 0$:** This is a property of the Vasicek conditional-PD function, whose value at $Z = 0$ does not equal the unconditional PD; the projection intercept is recentred so the baseline reproduces the through-the-cycle default rate. Under the report-wide convention that $Z < 0$ is adverse, the baseline therefore reads as mildly adverse even though it is the central expectation.
- **Categorical Feature Encoding:** Grade, term, employment length and home ownership are ordinal-encoded before WoE binning. Label ordering assumptions (e.g., A=1 through G=7) are reasonable but should be validated against observed default rates for each category.

10.5 EMPIRICAL RESULTS VS. PUBLISHED LITERATURE

TABLE 27. Project empirical results vs. published reference ranges (comparison against the literature, not a reproduction of the cited studies).

Metric	This Study	Published Range	Source	Verdict	Comment
Scorecard Gini (OOT)	0.3977	0.30 – 0.45	Lessmann et al. 2015	Within range	
LightGBM Gini (OOT)	0.4055	0.35 – 0.55	Lessmann et al. 2015	Within range	
Mean LGD (P2P)	0.8931	0.25 – 0.45	Bellotti and Crook 2012	Above typical range	The OOS-validated LightGBM severity model puts mean LGD above both cited bands, consistent with low post-charge-off recovery on unsecured instalment loans once EAD is principal-only; the rejected two-stage model materially under-predicted severity (OOS $R^2 \approx -2.70$)
Downturn LGD	1.0000	0.25 – 0.45	Bellotti and Crook 2012	Above typical range	Conservative 90th-percentile downturn uplift above the mean band
IFRS 9 ECL Coverage	25.302%	40% – 45%	European Banking Authority 2022	Below typical range	The published band is the EBA NPL coverage ratio (allowance on <i>non-performing</i> exposure only); the project figure is whole-book lifetime ECL over total EAD, which blends performing Stage 1 exposure and therefore sits below impaired-only coverage — not a like-for-like gap
IFRS 9 Stage 2%	29.4%	8% – 11%	European Banking Authority 2022	Above typical range	Stage 2 share well above the EBA EU-aggregate ($\approx 9\text{--}10\%$): expected for an unsecured, high-yield US consumer book with materially higher SICR incidence than the diversified EU bank averages

Verdicts in Table 27 are recomputed at every build by numerical comparison of the two adjacent columns; out-of-range values are reported as such, with the driver noted in the Comment column.

10.6 RECOMMENDED NEXT STEPS

- 1. Independent Validation:** Conduct an independent review of the binning algorithms and scorecard points scaling to verify math and logical consistency.
- 2. Advanced LGD Modeling:** Explore survival-based LGD models that capture time-varying recovery patterns over the life of defaulted assets.
- 3. Longitudinal SICR Tracking:** Implement a proper SICR framework using origination-date PD snapshots stored at booking, enabling precise Stage 2 migration rates compliant with IFRS 9 paragraph 5.5.9.

11 APPENDICES

11.1 A. SCORECARD SCALING DERIVATION

To map the log-odds predicted by the logistic regression to a scaled credit score, we solve the linear system:

$$\text{Score} = \text{Offset} + \text{Factor} \times \ln(\text{odds}) \quad (23)$$

Subject to the boundary conditions:

- Score = 600 at odds = 50 : 1 ($\ln(50) \approx 3.912$)
- Score = 620 at odds = 100 : 1 ($\ln(100) \approx 4.605$, satisfying PDO = 20)

Solving for the scaling parameters:

$$\text{Factor} = \frac{20}{\ln(100) - \ln(50)} = \frac{20}{\ln(2)} \approx 28.8539 \quad (24)$$

$$\text{Offset} = 600 - 28.8539 \times \ln(50) \approx 487.123 \quad (25)$$

11.2 B. BASEL IRB “OTHER RETAIL” SUPERVISORY PARAMETERS

Under BCBS §322–328, the regulatory correlation (R) is constrained between 0.03 and 0.16. The correlation decays exponentially as the PD increases, reflecting the lower correlation of defaults among higher-risk borrowers during stable economic conditions:

$$R = 0.03 \times \frac{1 - e^{-35 \times PD}}{1 - e^{-35}} + 0.16 \times \left[1 - \frac{1 - e^{-35 \times PD}}{1 - e^{-35}} \right] \quad (26)$$

For the RWA calculation, the supervisory PD floor of **0.03%** is strictly applied to ensure a minimum capital charge.

11.3 C. TECHNICAL STACK & REPRODUCIBILITY

The pipeline is designed to be fully reproducible:

TABLE 28. Modeling Pipeline Technology Stack

Component	Technology	Purpose in Pipeline
Language	Python 3.11+	Core language and scripting
PD Underwriting	statsmodels Logit	Monotonic credit scorecard fitting
LGD Modeling	LightGBM (deployed); statsmodels GLM	Severity model selected out-of-sample; the two-stage cure + fractional-logit model is the benchmarked <i>challenger</i> that was not deployed (Section 4.2)
Challenger Model	LightGBM	Non-linear machine learning benchmark
Survival Curves	PD Discrete-time hazard model	Monthly lifetime term structure modeling
PDF Generation	XeLaTeX + biber	Publication-quality academic PDF

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